

Encoding and Modulating

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Computer Engineering Department

Kasetsart University, Bangkok, Thailand

Outline

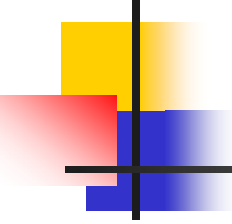
- **Digital-to-digital** conversion
 - Encode digital data into a digital signal
 - Sending computer data
- **Analog-to-Digital** conversion
 - Digitizing an analog
 - Sending voice in telephone
(Decrease effect of noise)



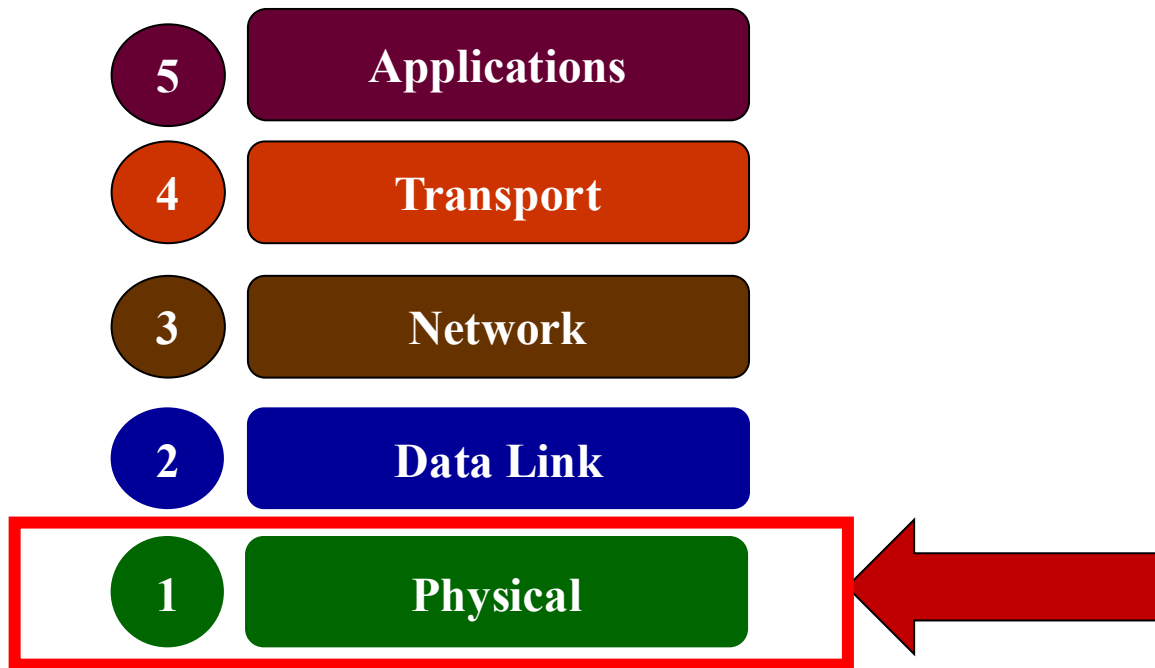
Outline

- **Digital-to-Analog** conversion
 - Modulating a digital signal
 - Sending computer data through public telephone line
- **Analog-to-Analog** conversion
 - Modulating an analog signal
 - Sending music from radio station



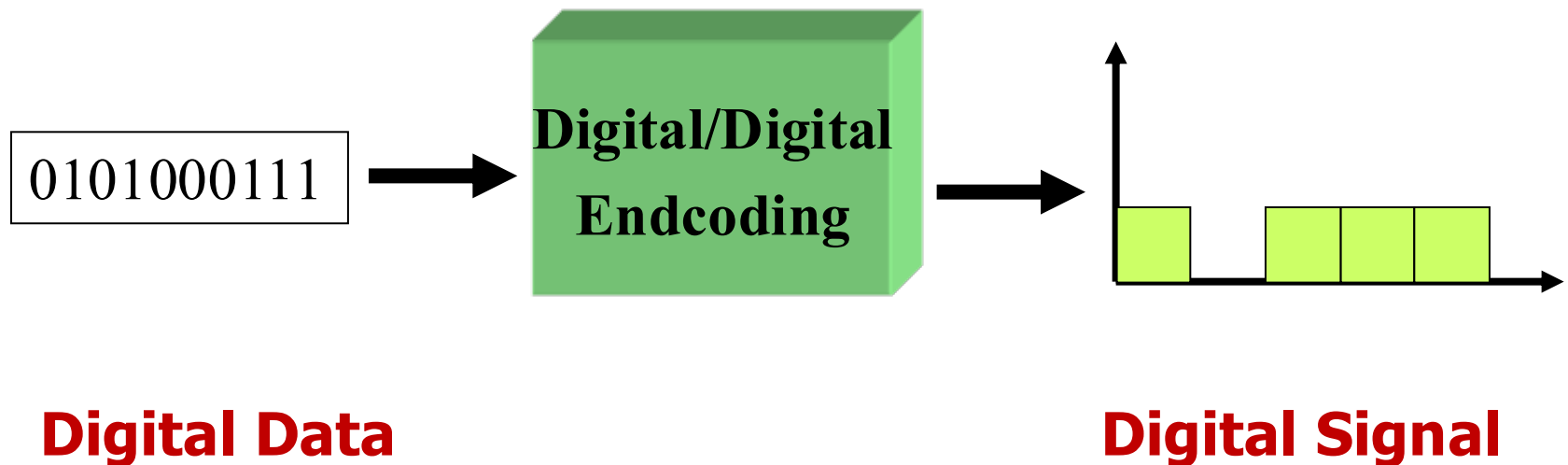


Encoding and Modulating



Digital-to-Digital conversion

- Encoding data into a digital signal



Digital to Digital Conversion

Unipolar

Polar

Bipolar

NRZ

RZ

Biphase

AMI

B8ZS

HD3B

NRZ-L

NRZ-1

Manchester

Diff-Manc

Digital to Digital Conversion

```
graph TD; Root[Digital to Digital Conversion] --> Unipolar[Unipolar]; Root --> Polar[Polar]; Root --> Bipolar[Bipolar]; Polar --> NRZ[NRZ]; Polar --> RZ[RZ]; Polar --> Biphase[Biphase]; NRZ --> NRZ_L[NRZ-L]; NRZ --> NRZ_1[NRZ-1]; Biphase --> Manchester[Manchester]; Biphase --> Diff_Manc[Diff-Manc]; Bipolar --> AMI[AMI]; Bipolar --> B8ZS[B8ZS]; Bipolar --> HD3B[HD3B];
```

Unipolar

Polar

Bipolar

NRZ

RZ

Biphase

AMI

B8ZS

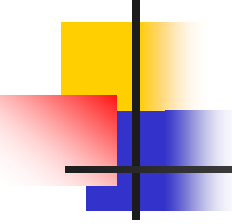
HD3B

NRZ-L

NRZ-1

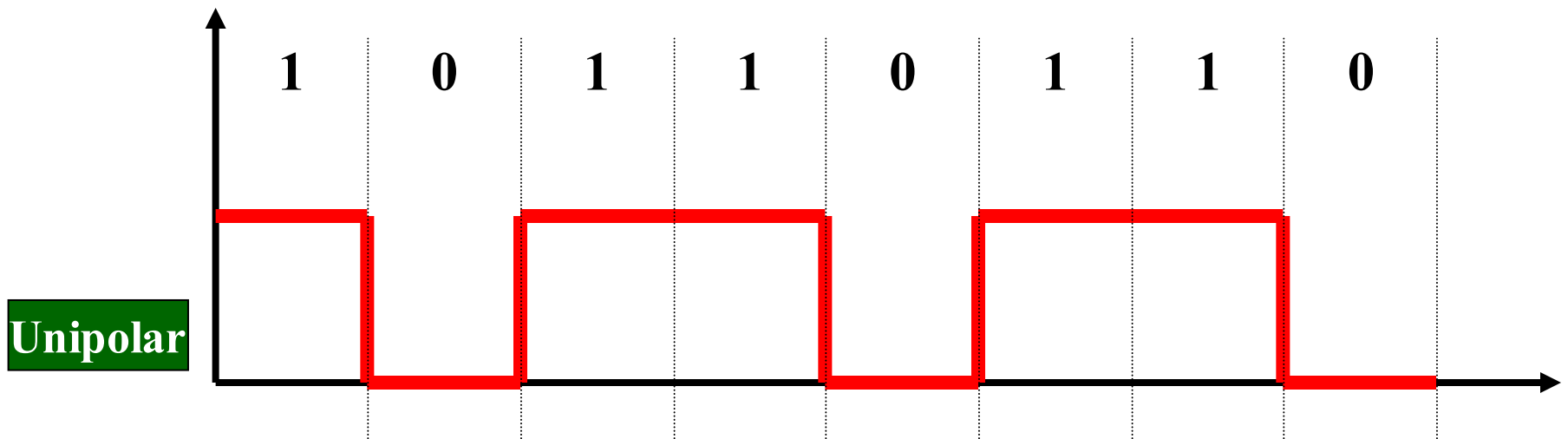
Manchester

Diff-Manc



Unipolar Encoding

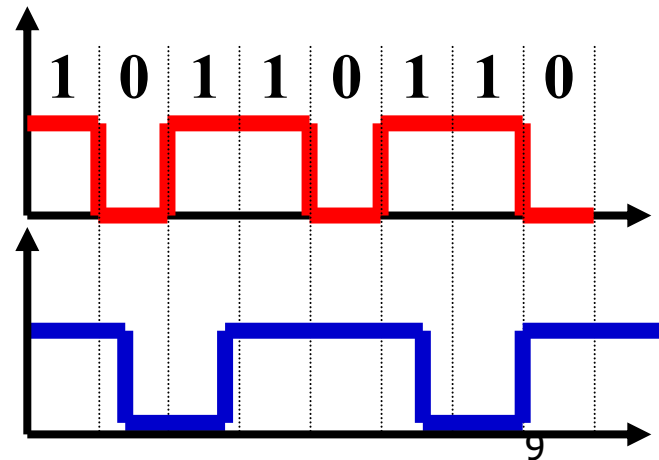
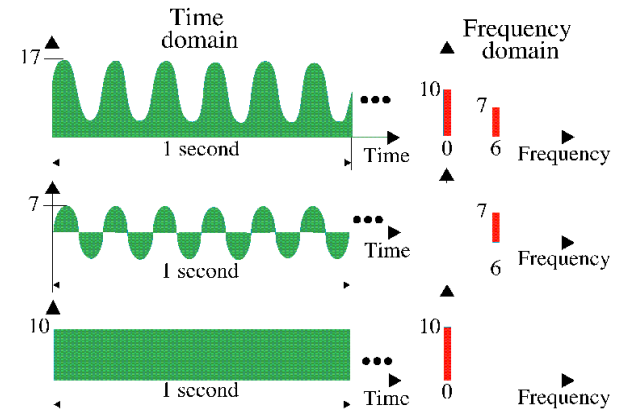
- One level of value



Unipolar Encoding

■ Problems

- DC components
 - Cannot pass through some media
- Synchronization
 - Beginning/ending problem (1111111111)
 - Distortion (four 1111 → five 1111)
 - Solved by separate line



Digital to Digital Conversion

```
graph TD; Root[Digital to Digital Conversion] --> Unipolar[Unipolar]; Root --> Polar[Polar]; Root --> Bipolar[Bipolar]; Polar --> NRZ[NRZ]; Polar --> RZ[RZ]; Polar --> Biphase[Biphase]; NRZ --> NRZ_L[NRZ-L]; NRZ --> NRZ_1[NRZ-1]; Biphase --> Manchester[Manchester]; Biphase --> Diff_Manc[Diff-Manc]; Bipolar --> AMI[AMI]; Bipolar --> B8ZS[B8ZS]; Bipolar --> HD3B[HD3B];
```

Unipolar

Polar

Bipolar

NRZ

RZ

Biphase

AMI

B8ZS

HD3B

NRZ-L

NRZ-1

Manchester

Diff-Manc

Polar Encoding

Two levels (+ and -)

NRZ (Nonreturn to Zero)

- Signal Level → state of bit

- Long Steam of 0 or 1

0 0 1 0 0 1 1 0

NRZ-L

- Inverted Signal Level → if “1”

- Long Steam of 0 + Syschronize @ 1

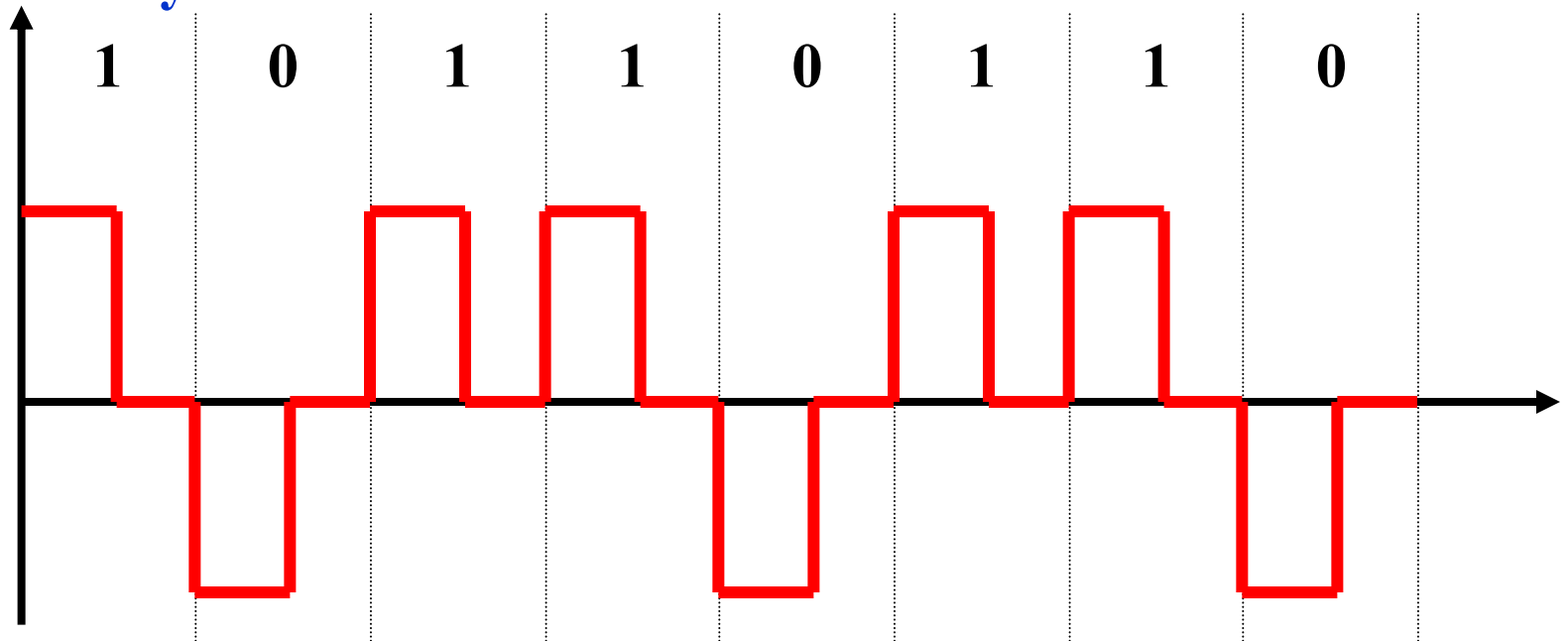
NRZ-I

Polar Encoding

RZ (Return to Zero)

- Three levels (+ - 0)
 - 2 signal changes per bit → more BW

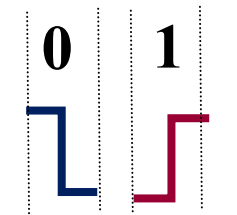
+ Synchronization



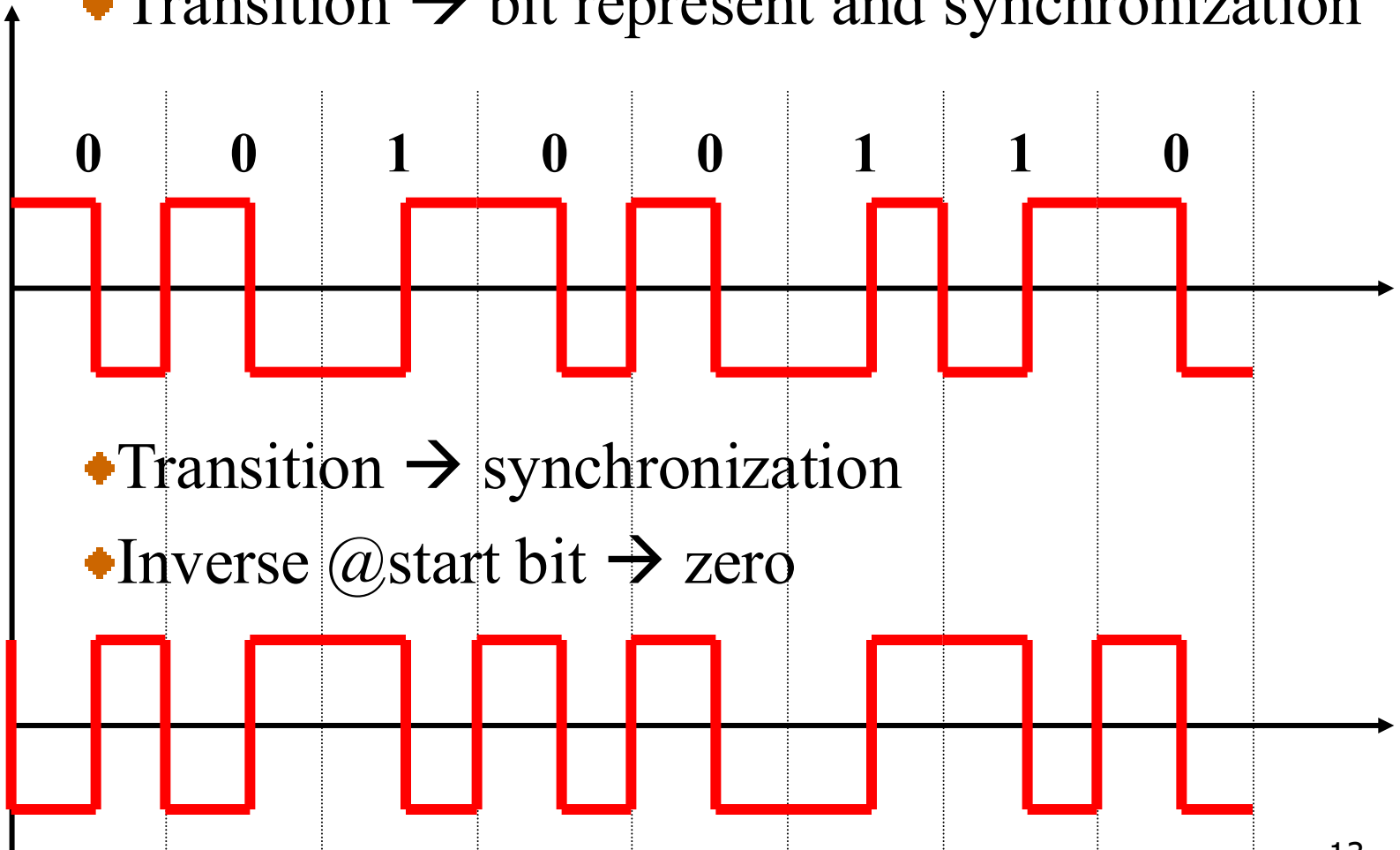
Polar Encoding

Biphase

♦ Transition → bit represent and synchronization



Manchester

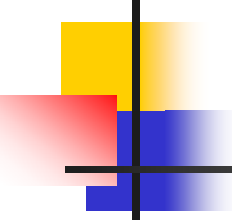


♦ Transition → synchronization

♦ Inverse @start bit → zero

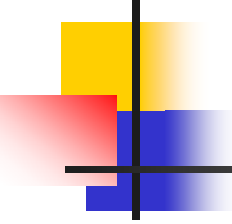
0 – return
to same
1 – opposite

Differential
Manchester



Problems

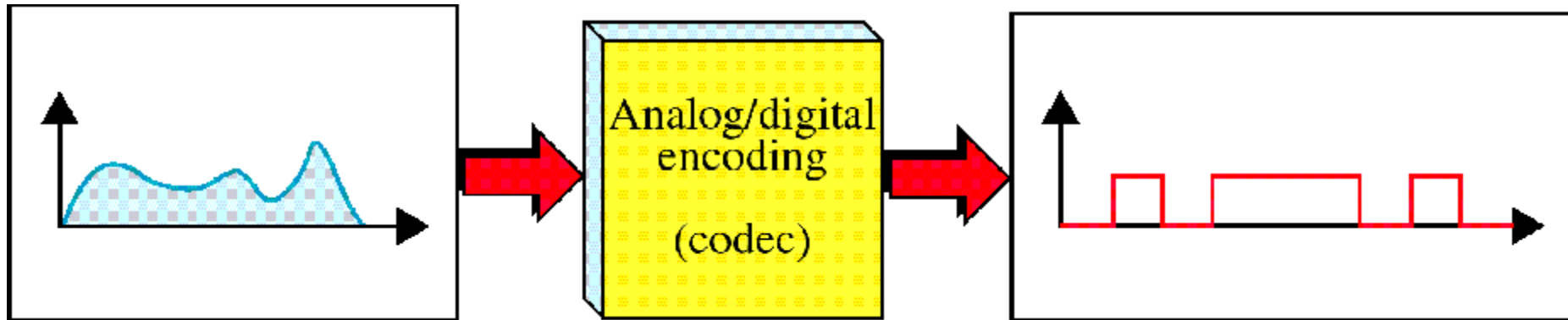
- DC component
- Synchronization (Long stream of 0 or 1)



Outline

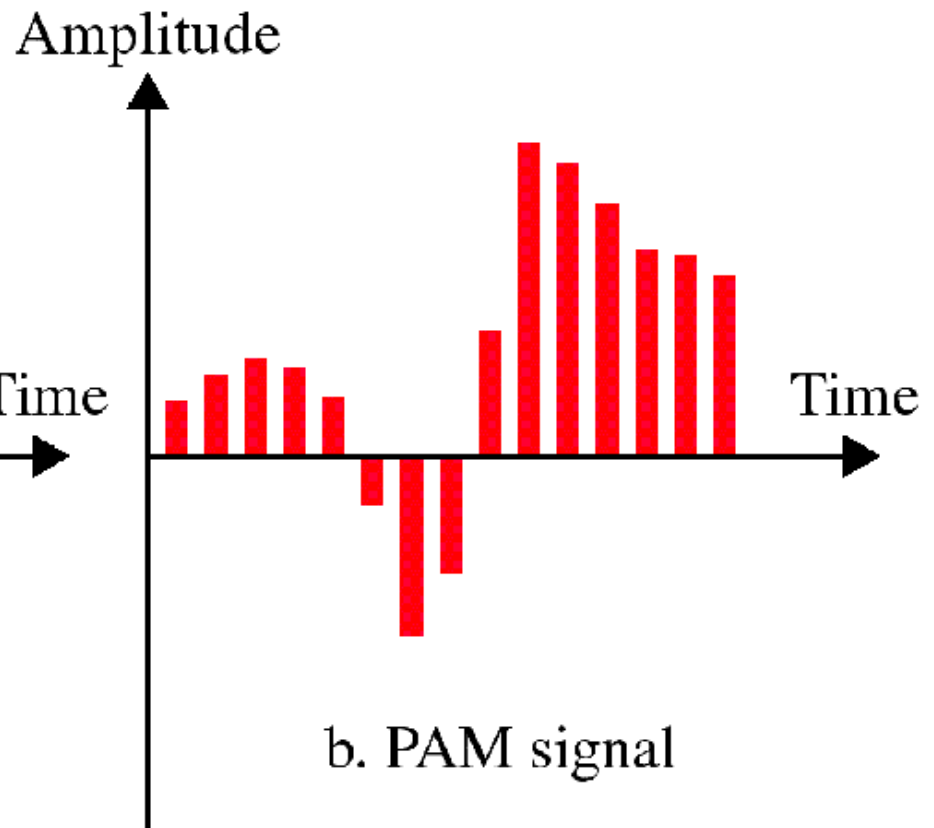
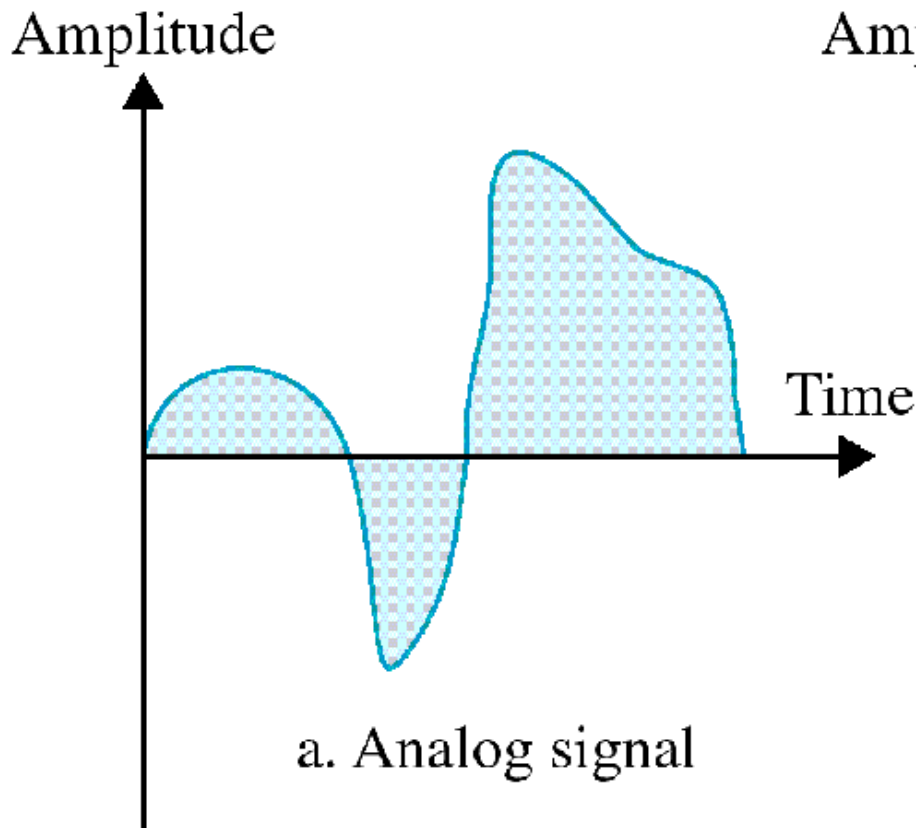
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Analog to Digital Encoding

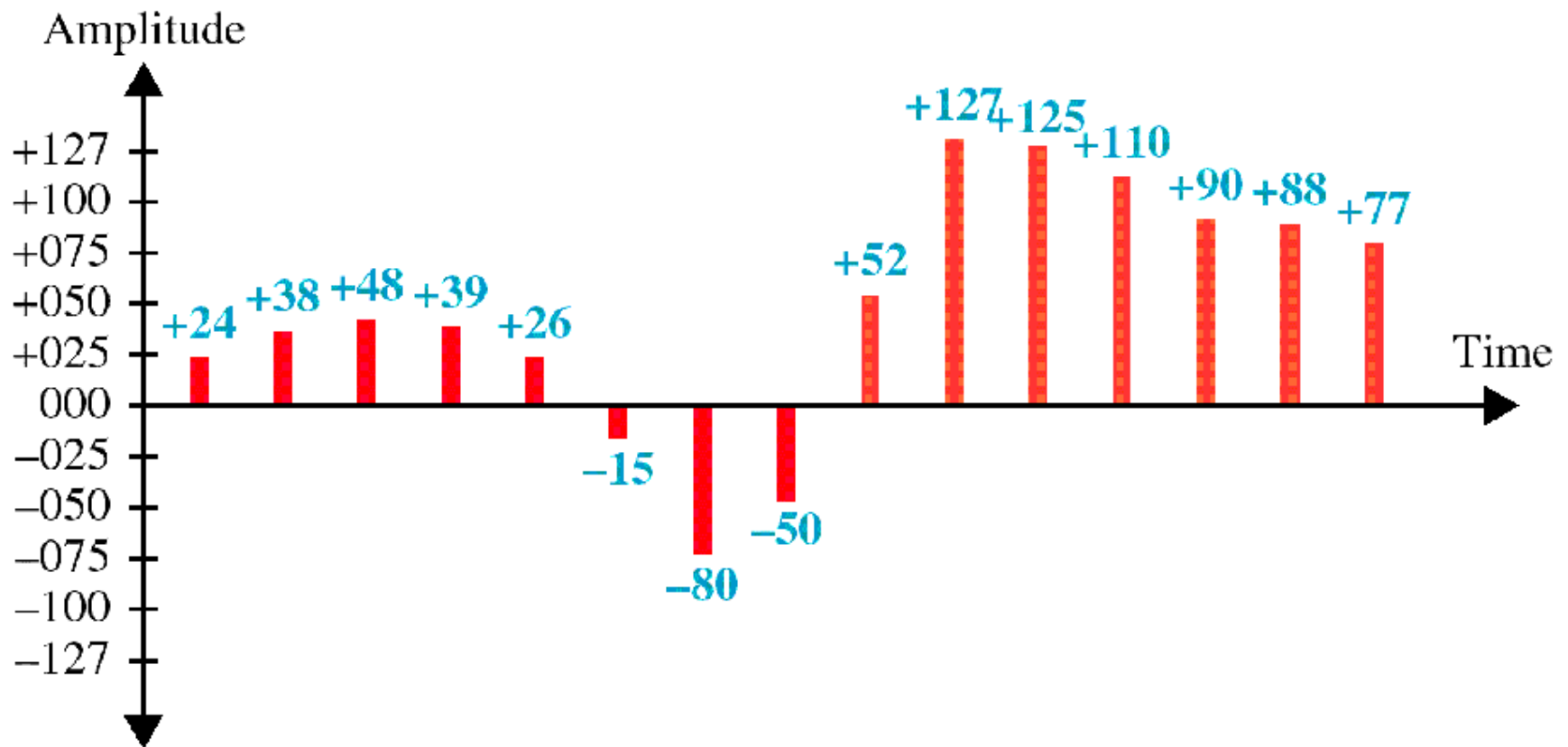


Step 1:

Pulse Amplitude Modulation(PAM)



Step 2: Quantized PAM Signal



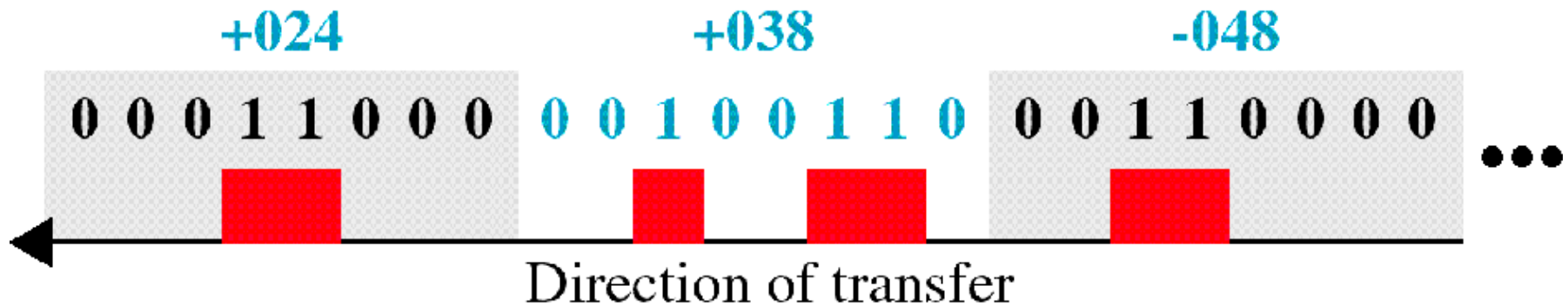
Step 3:

Quantizing Using Sign and Magnitude

+024	00011000	-015	10001111	+125	01111101
+038	00100110	-080	11010000	+110	01101110
+048	00110000	-050	10110010	+090	01011010
+039	00100111	+052	00110110	+088	01011000
+026	00011010	+127	01111111	+077	01001101

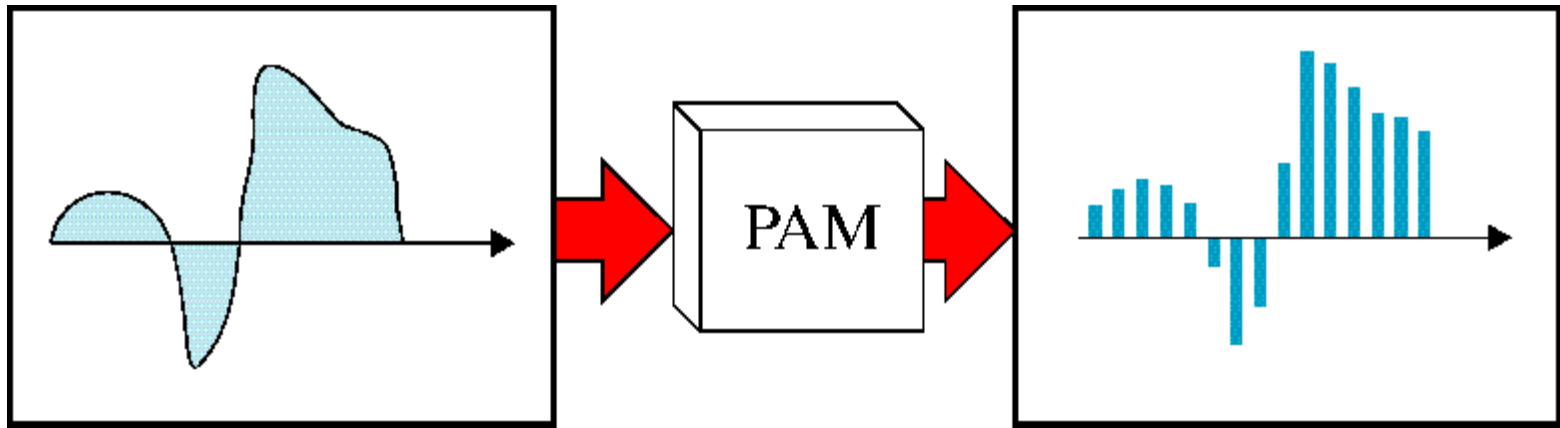
Sign bit
+ is 0 - is 1

Step 4: Pulse Code Modulation (PCM)

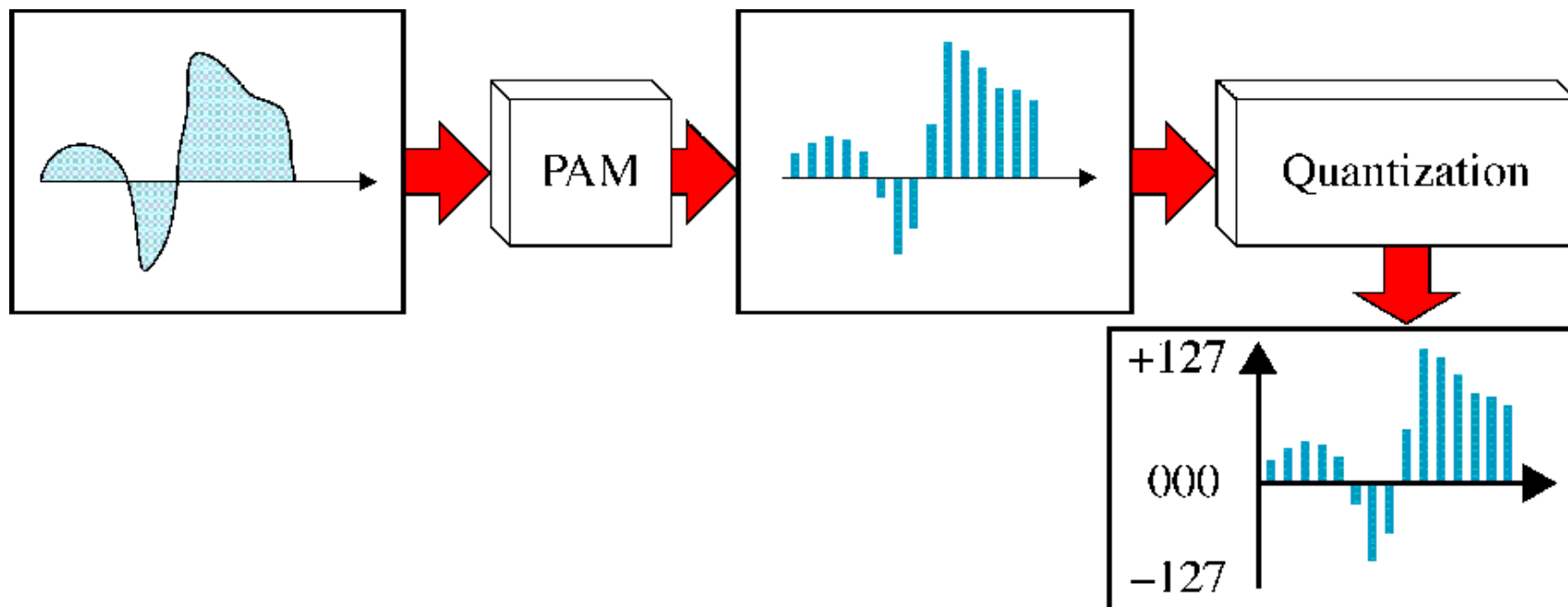


Summary

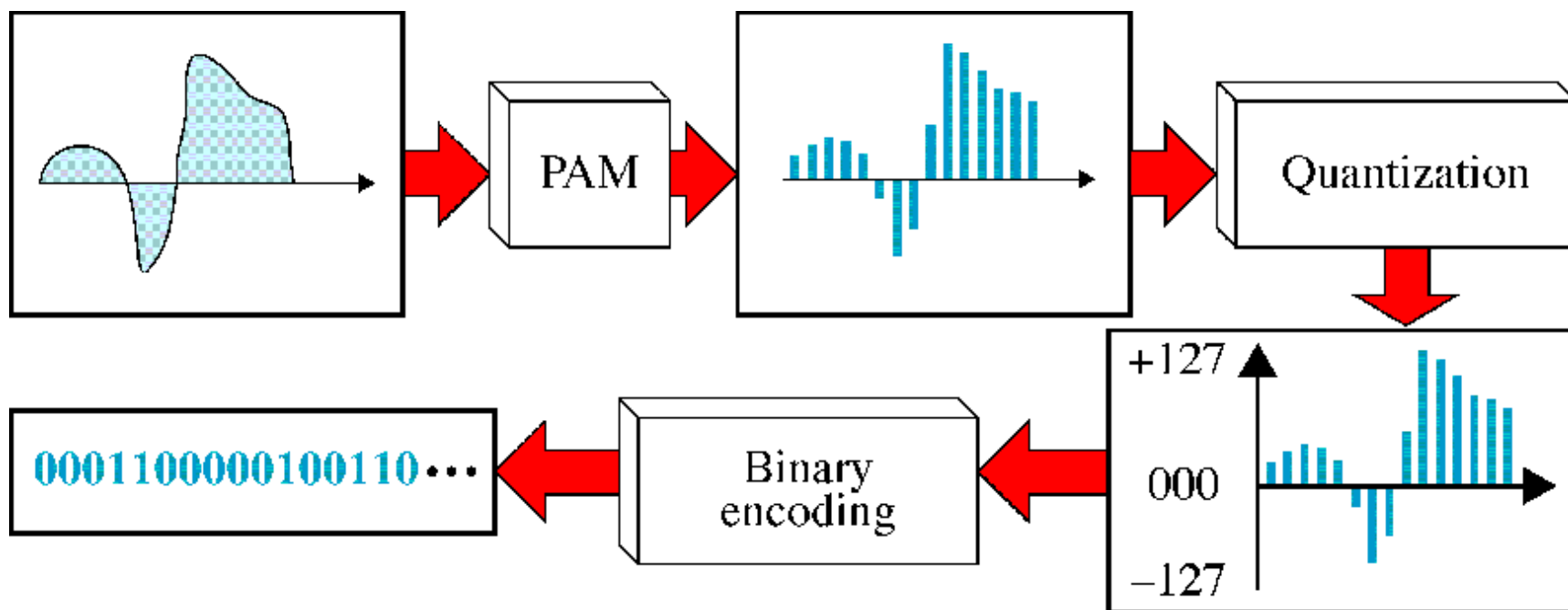
From Analog to PCM



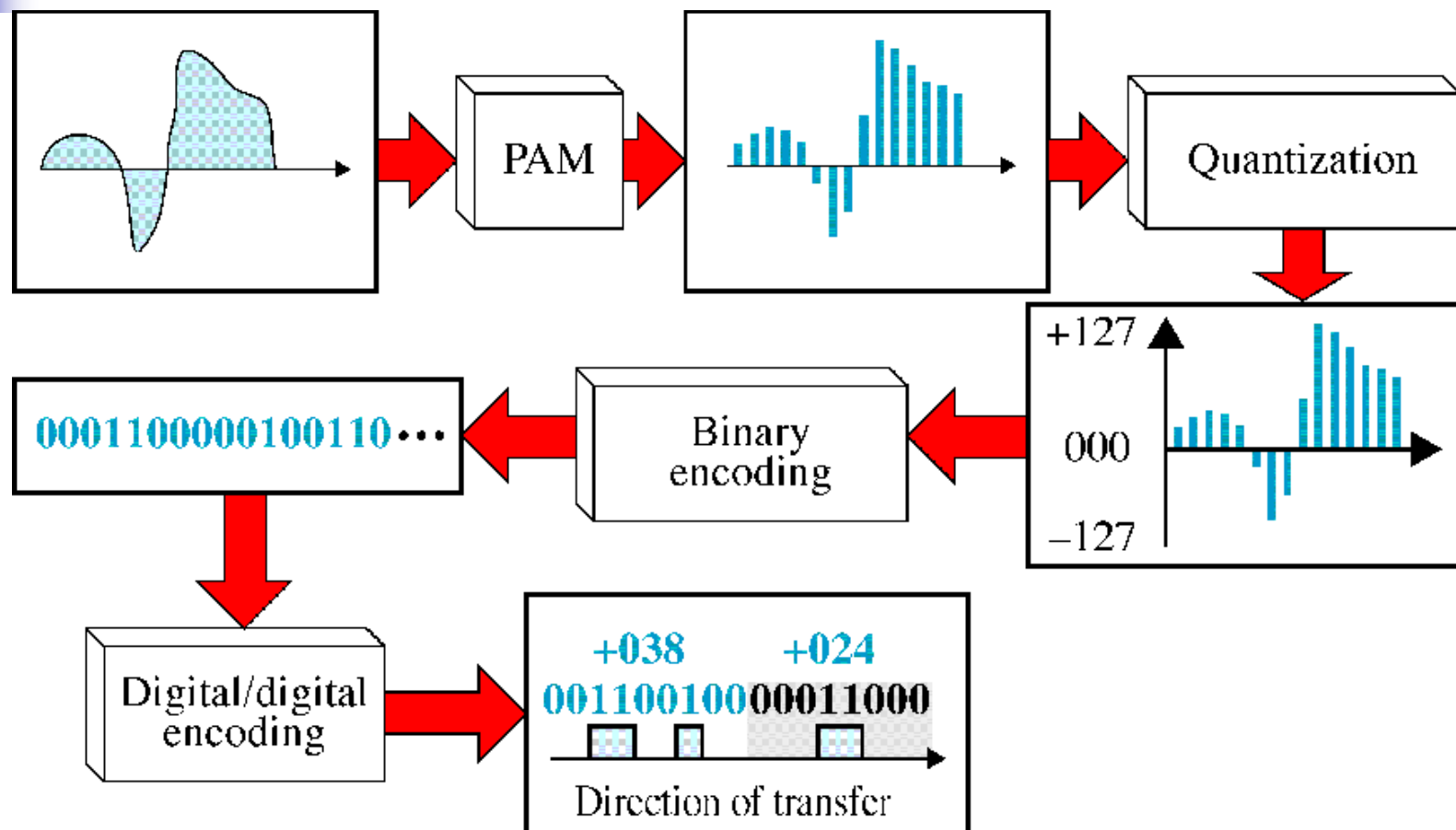
From Analog to PCM



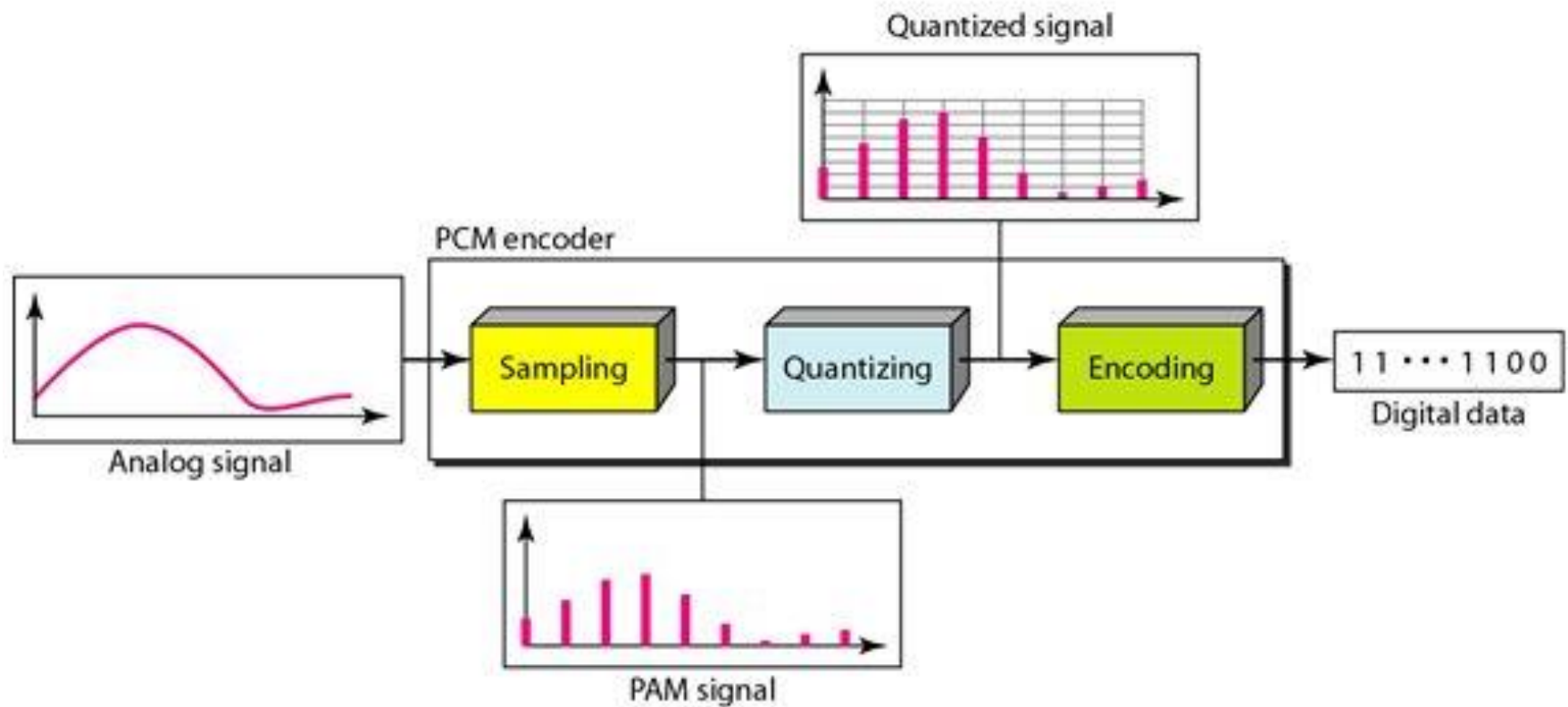
From Analog to PCM



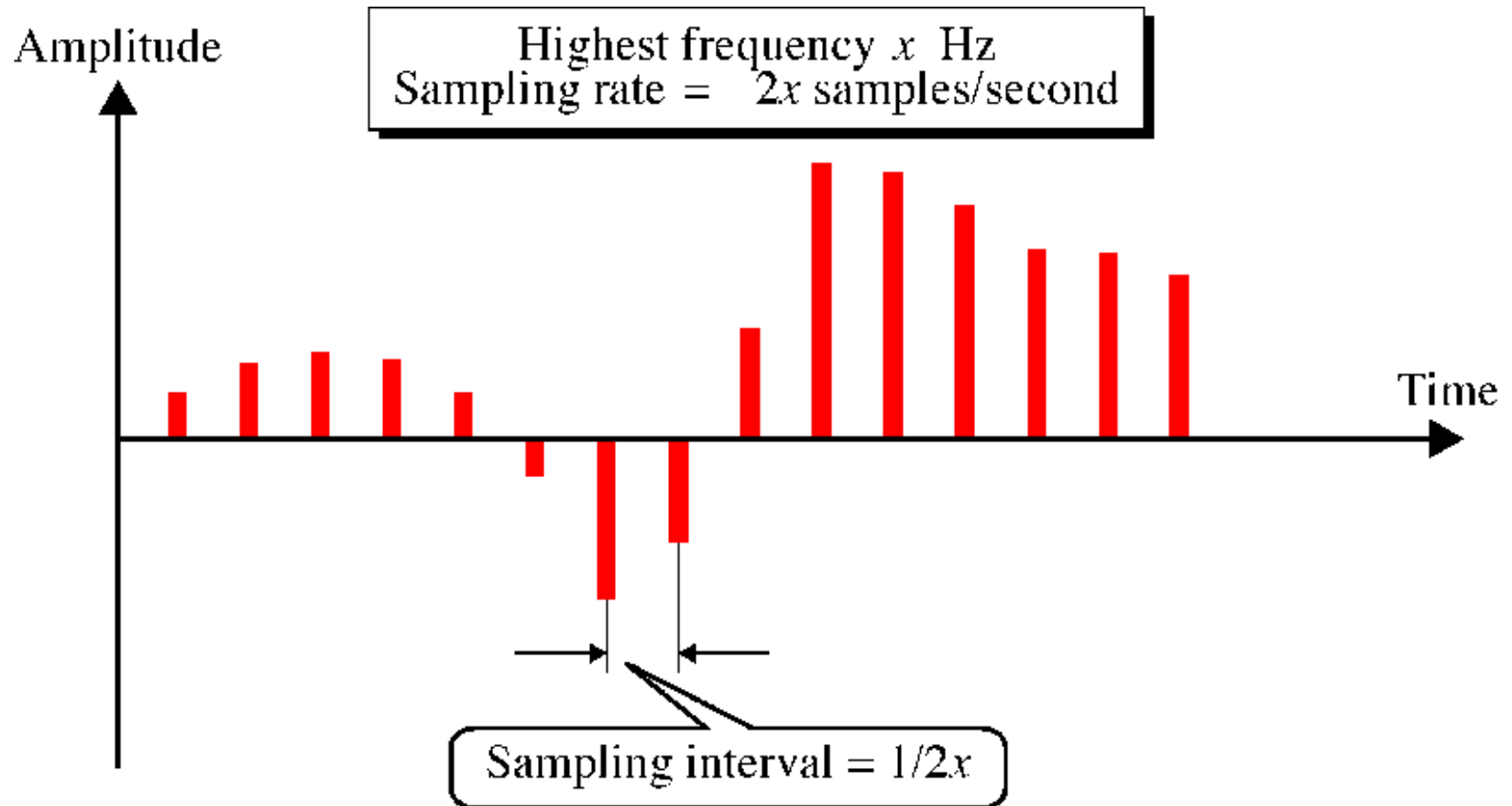
From Analog to PCM

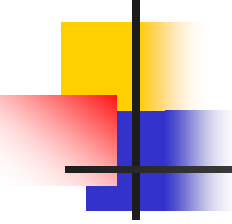


Analog to Digital Encoding



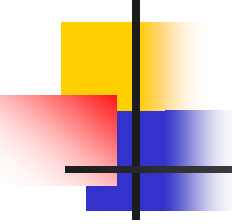
Nyquist Theorem





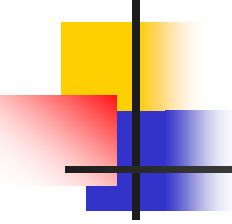
Outline

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Basic Concepts

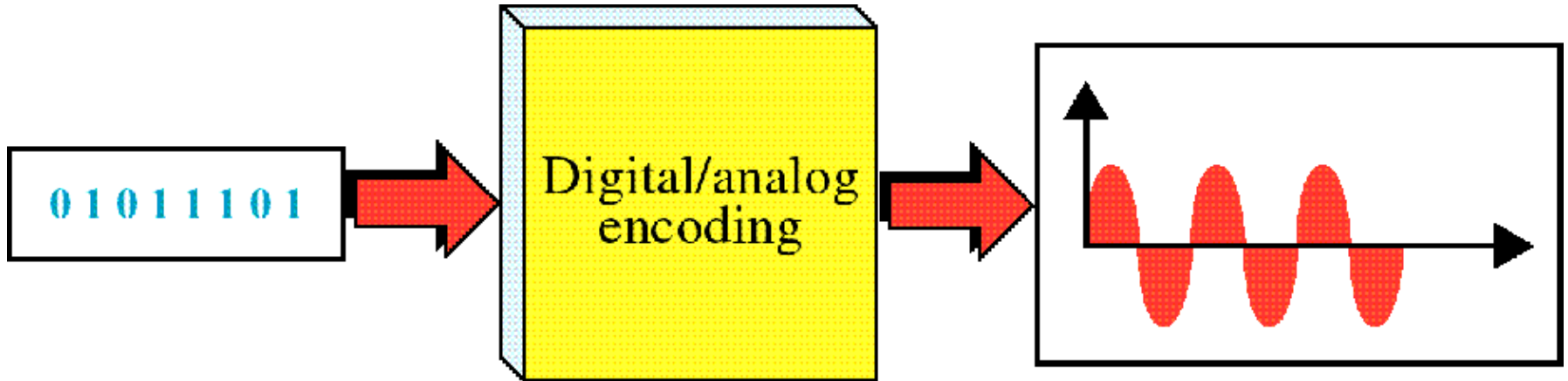
- Bit rate
 - Bits transfer in a second
 - Computer Efficiency (time to send)
- Baud rate
 - # of Signal per second
 - Data transmission (move data)
 - Few signal → efficiency
- Analogy (transportation)

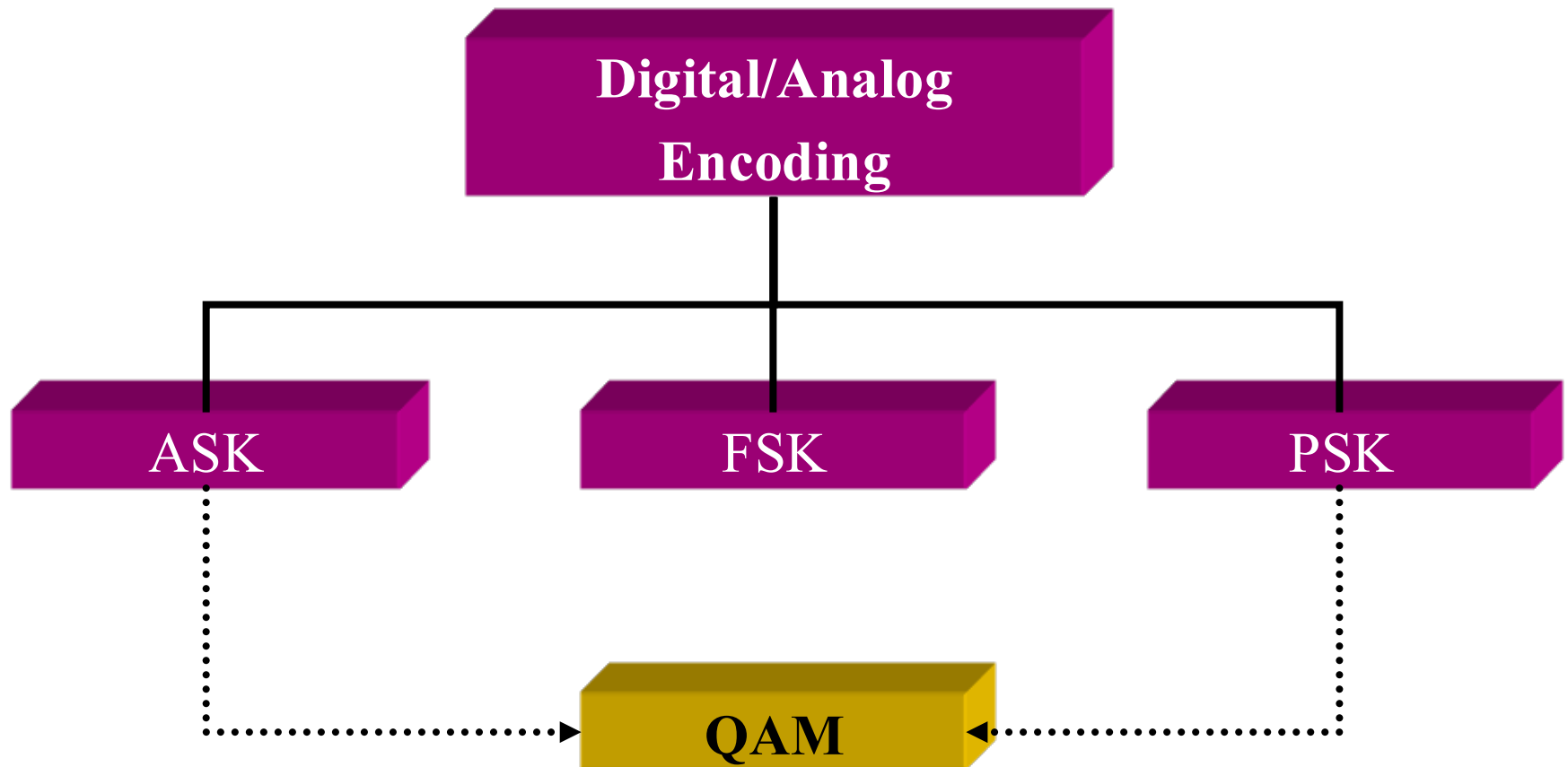


Basic concepts

- **Carrier signal (carrier frequency)**
 - High frequency as a basis for information
 - Sender and receiver agree on the frequency
 - Digital data is modulated (shift keying) on the carrier by modifying carrier characteristics
- **3 characteristics of carrier signal**
 - Amplitude
 - Frequency
 - Phase

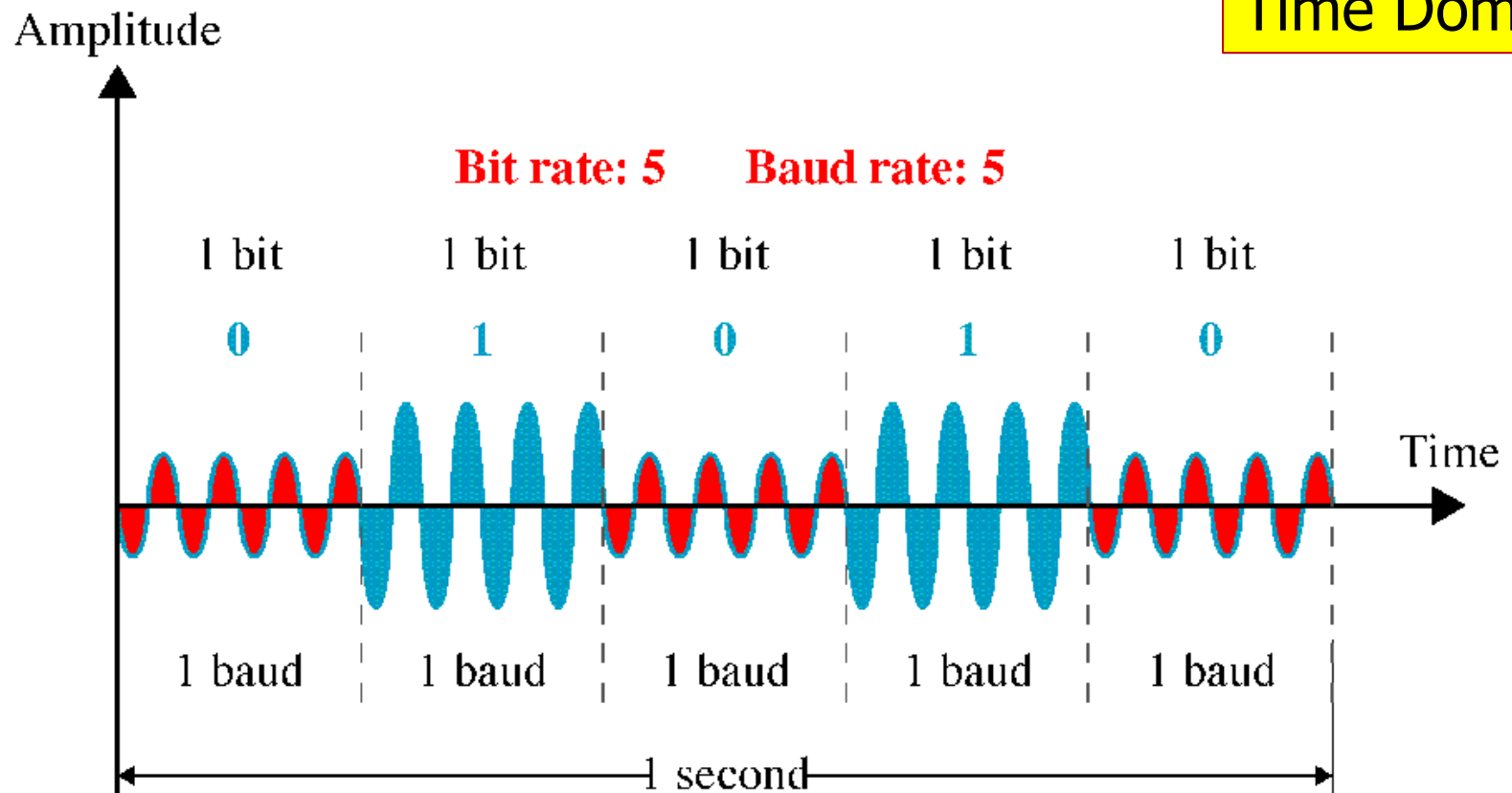
Digital to Analog Encoding





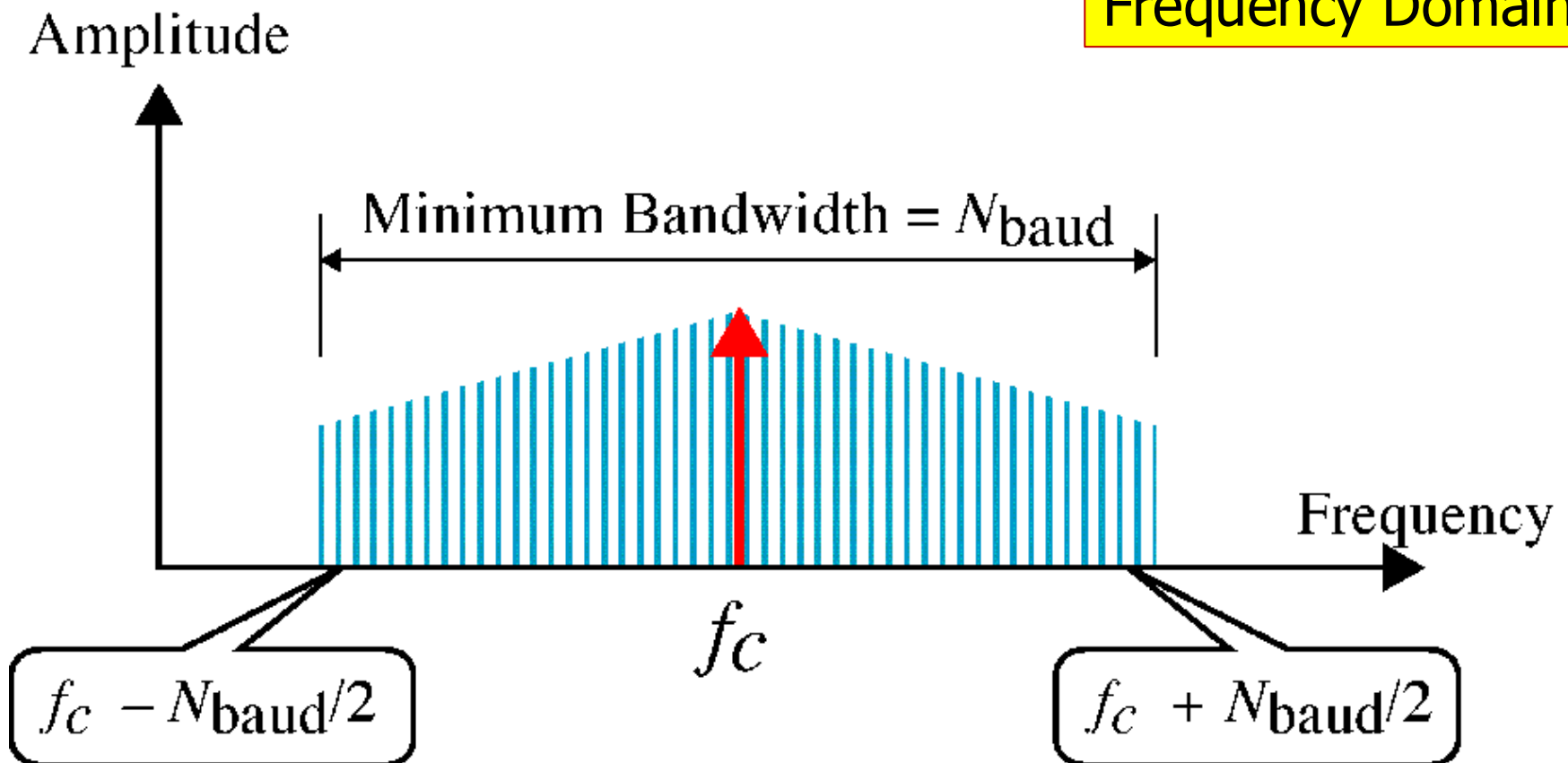
Amplitude Shift Keying (ASK)

Time Domain



Bandwidth for ASK

Frequency Domain

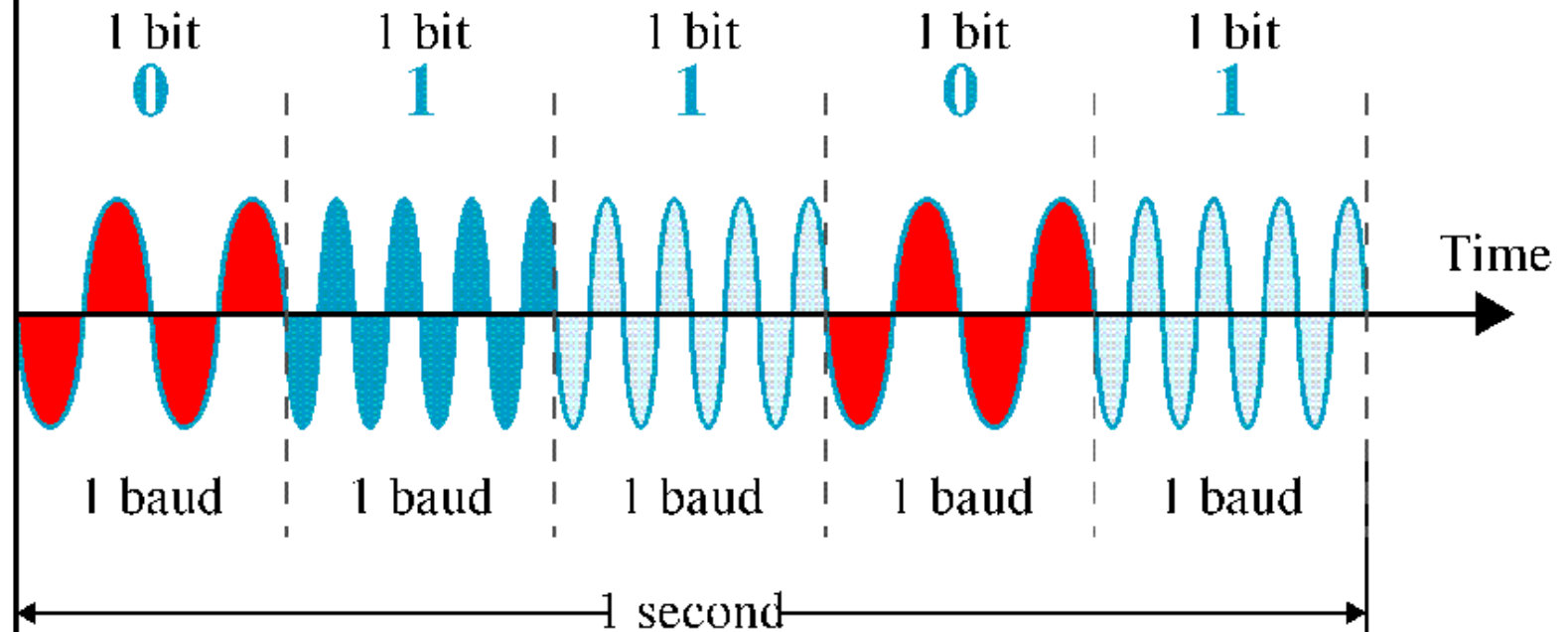


Frequency Shift Keying (FSK)

Time Domain

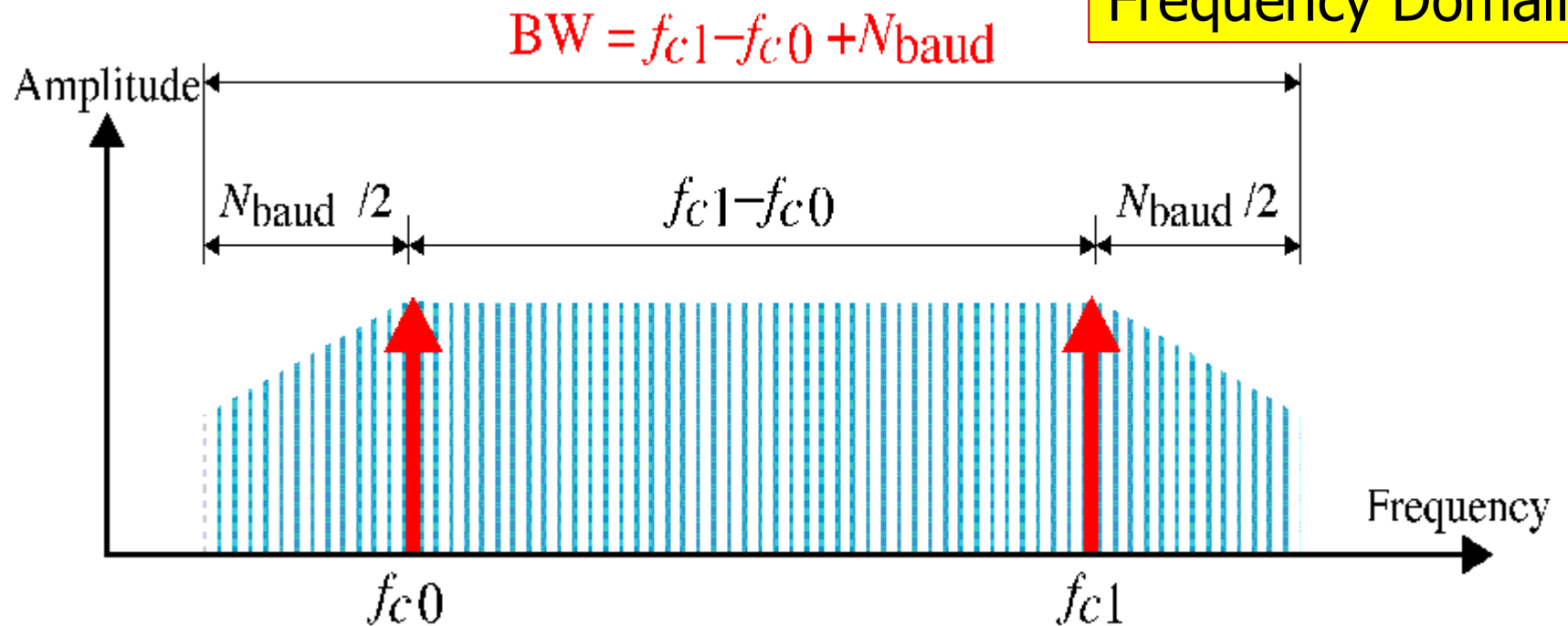
Amplitude

Bit rate: 5 Baud rate: 5



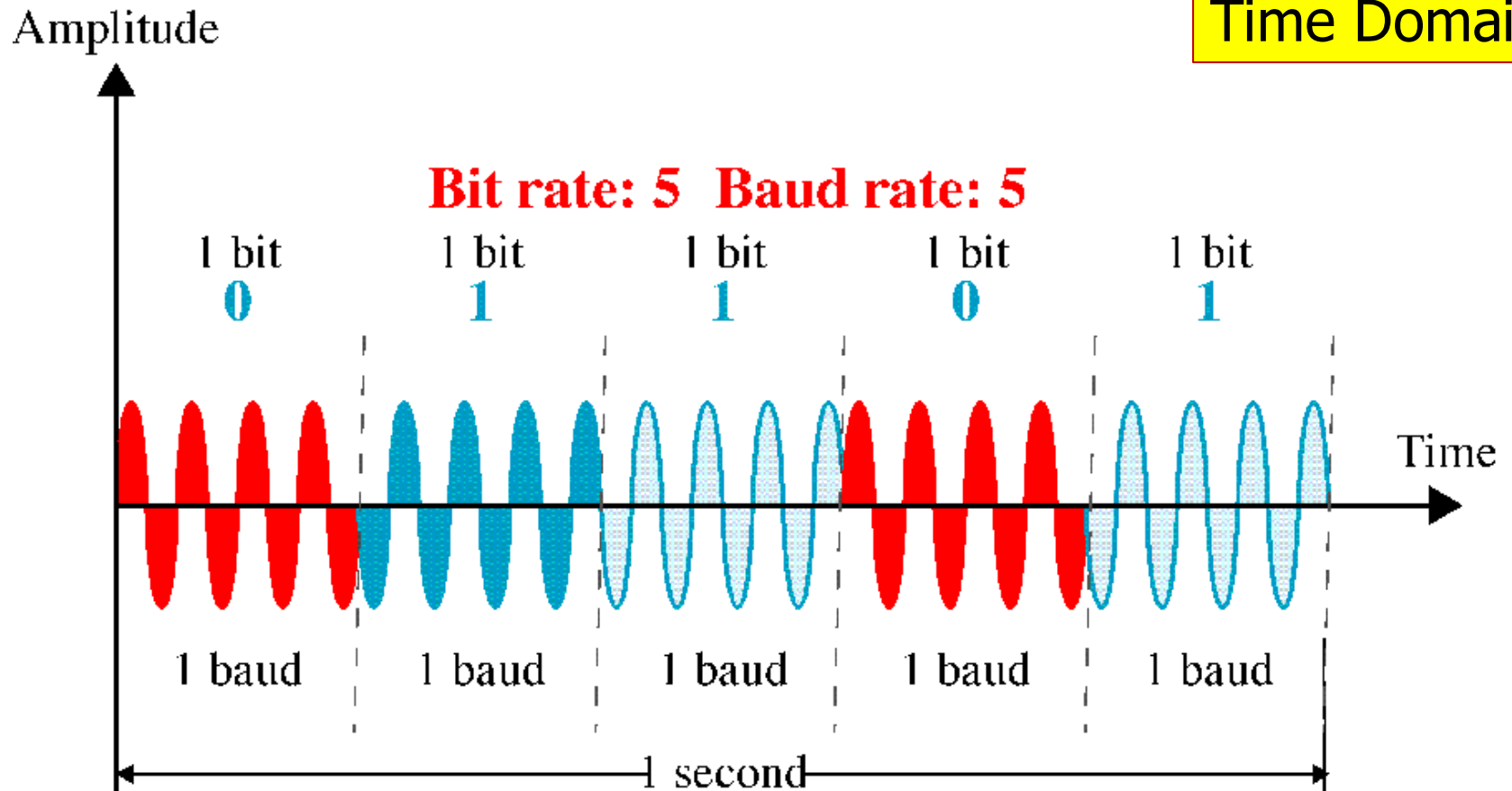
Bandwidth for FSK

Frequency Domain



Phase Shift Keying (PSK)

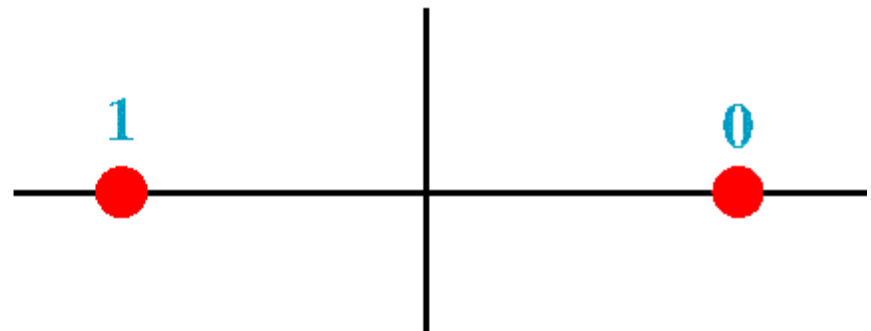
Time Domain



PSK Constellation

Bit	Phase
0	0
1	180

Bits

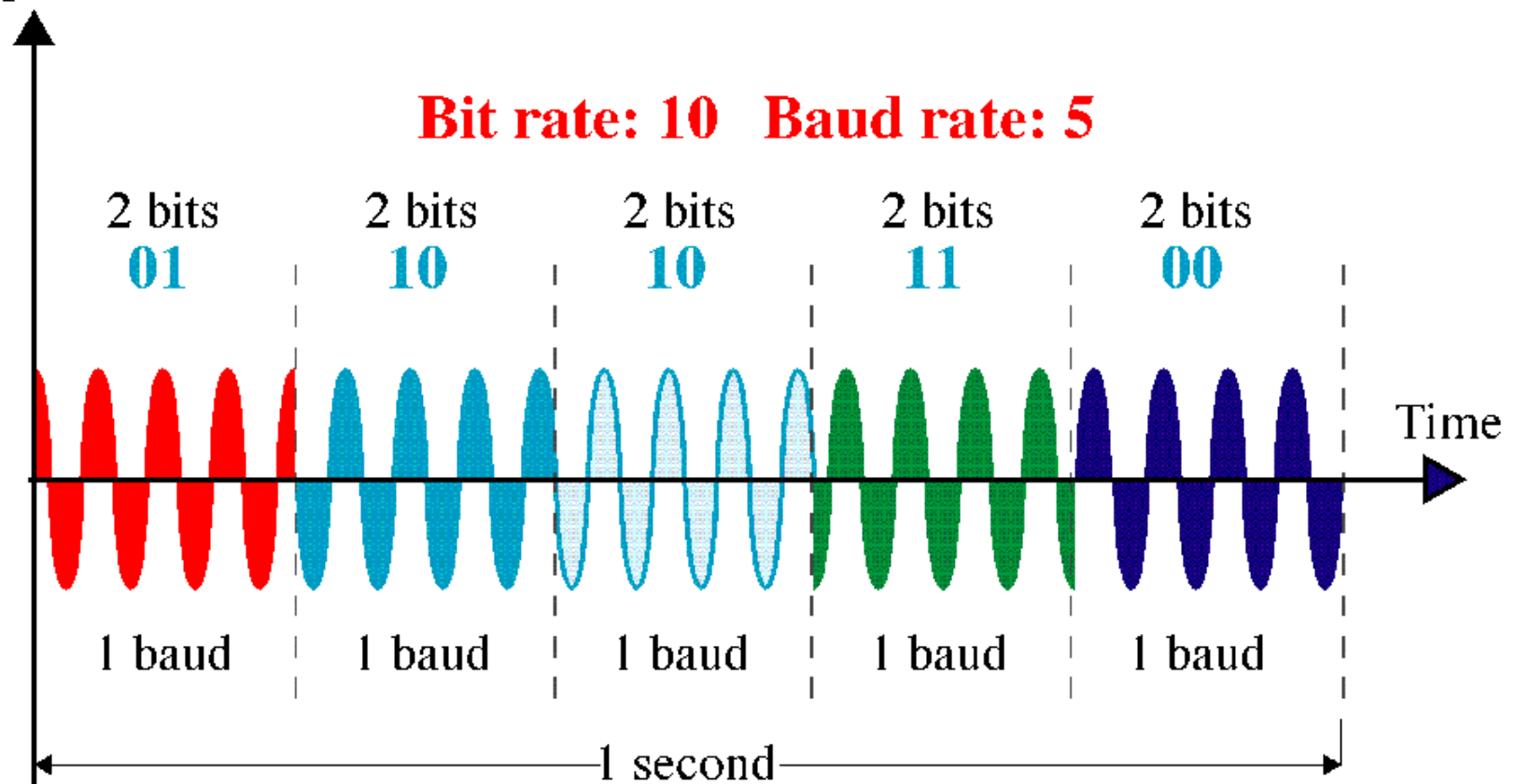


Constellation diagram

4-PSK

Time Domain

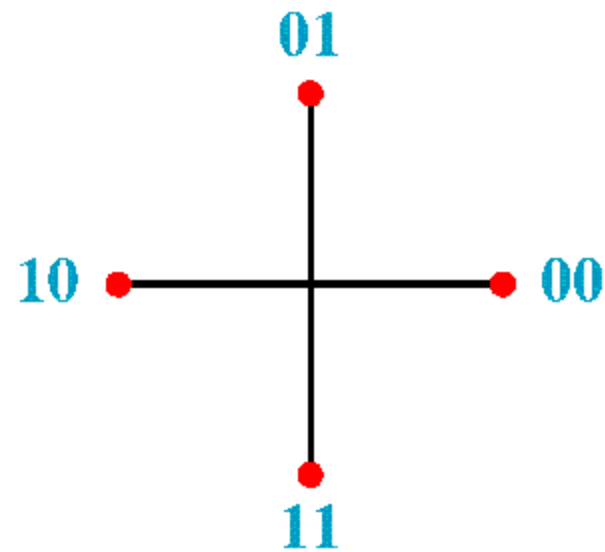
Amplitude



4-PSK Constellation

Dibit	Phase
00	0
01	90
10	180
11	270

Dibit
(2 bits)

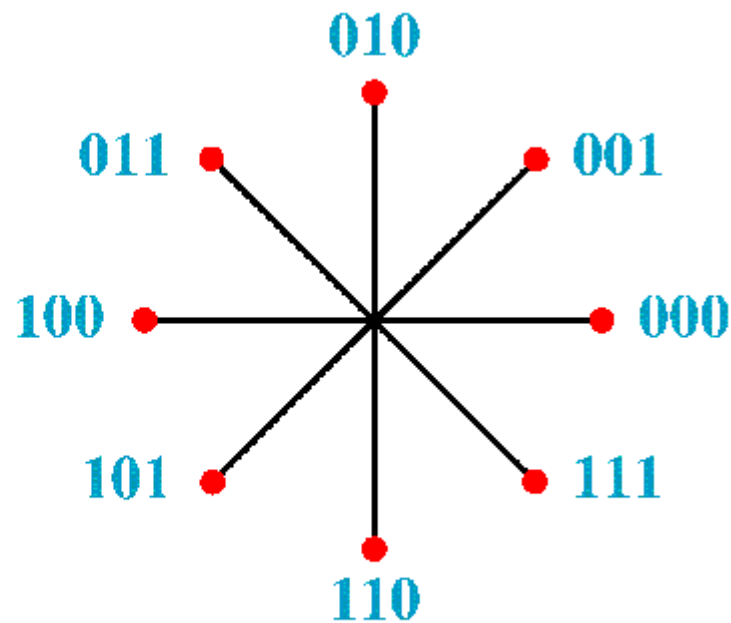


Constellation diagram

8-PSK Constellation

Tribit	Phase
000	0
001	45
010	90
011	135
100	180
101	225
110	270
111	315

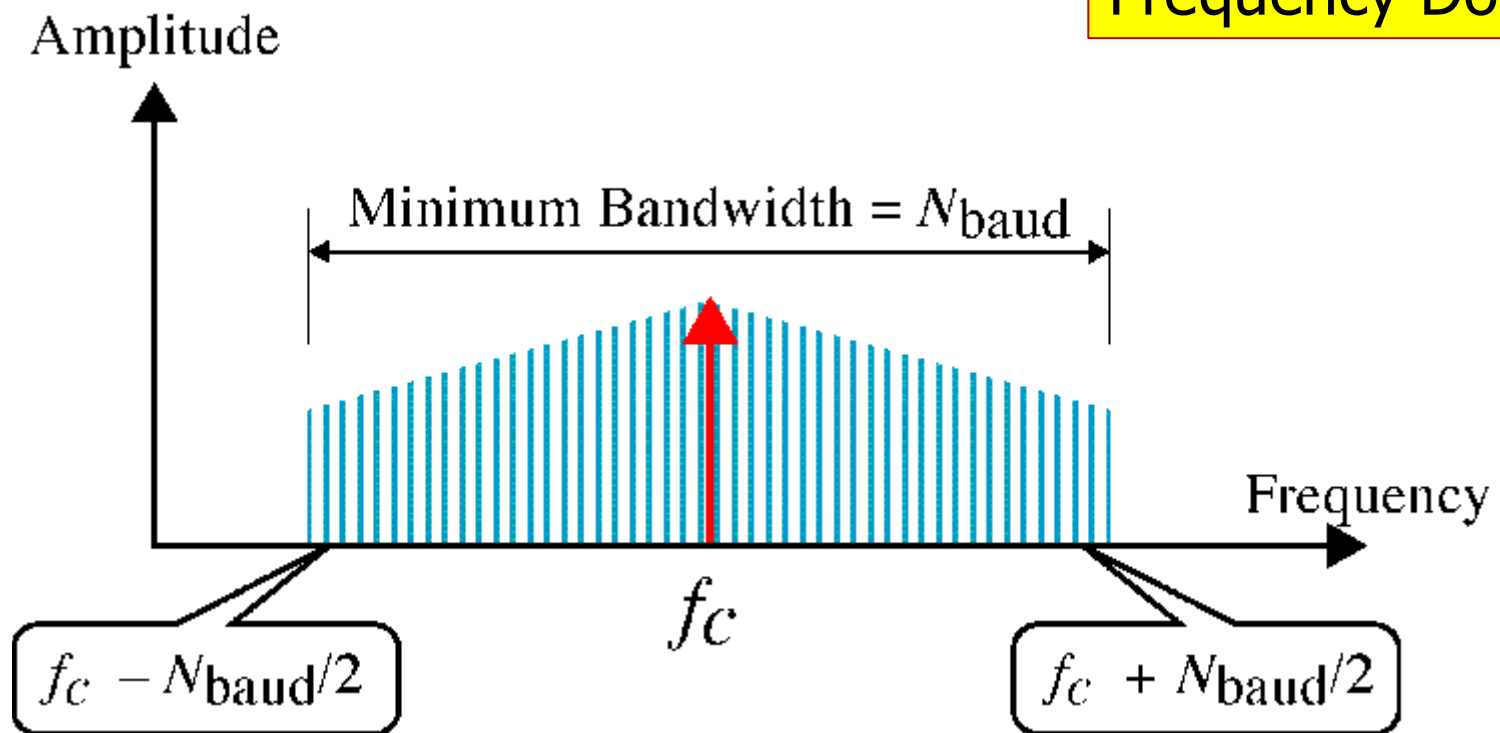
Tribits
(3 bits)



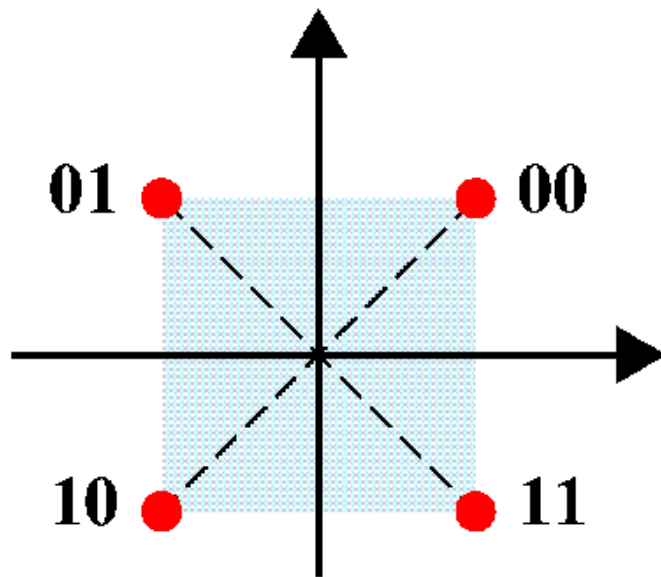
Constellation diagram

PSK Bandwidth

Frequency Domain

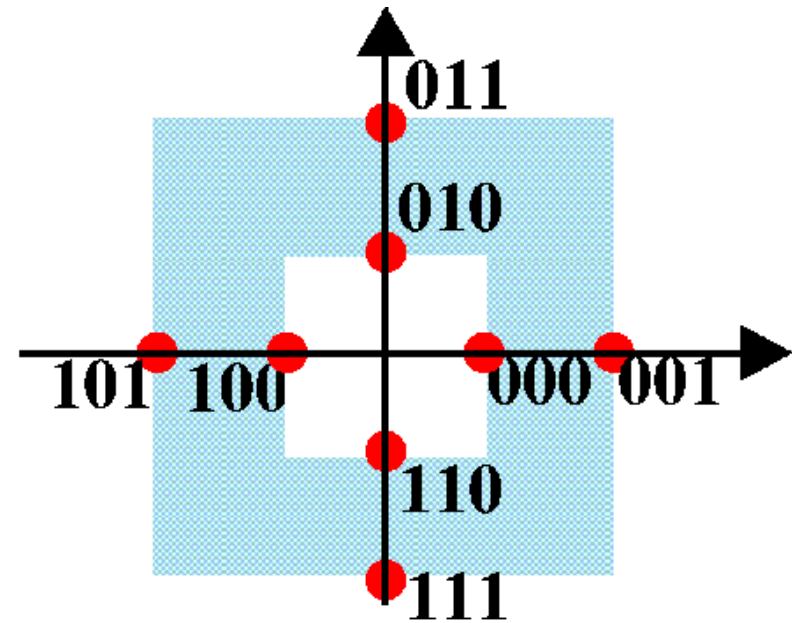


Quadrature Amplitude Modulation (QAM)



4-QAM

1 amplitude, 4 phases

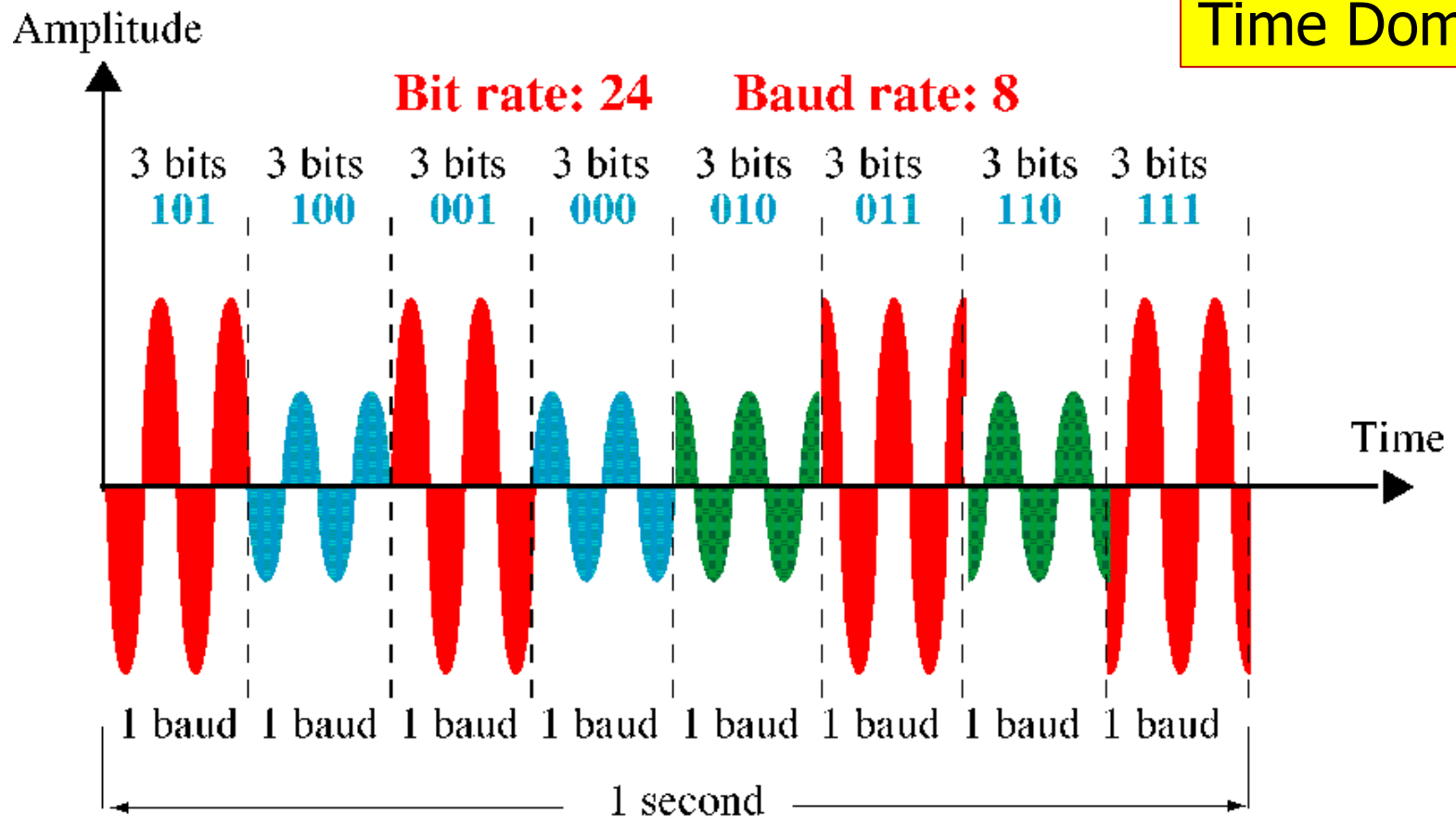


8-QAM

2 amplitudes, 4 phases

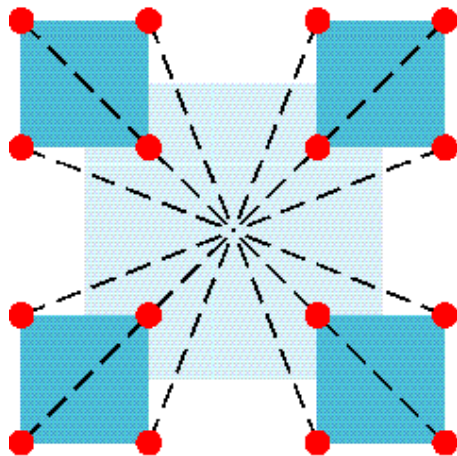
8-QAM Signal

Time Domain



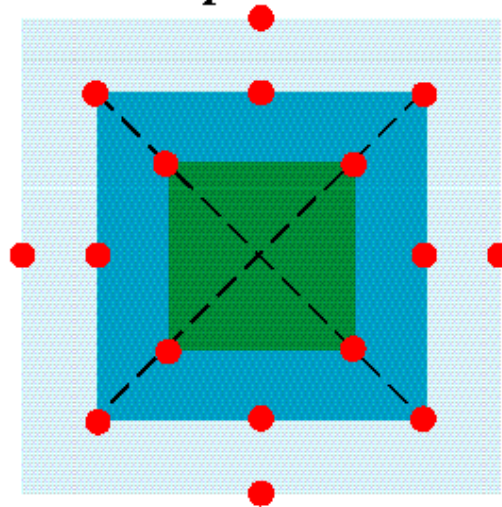
16-QAM Constellation

3 amplitudes,
12 phases



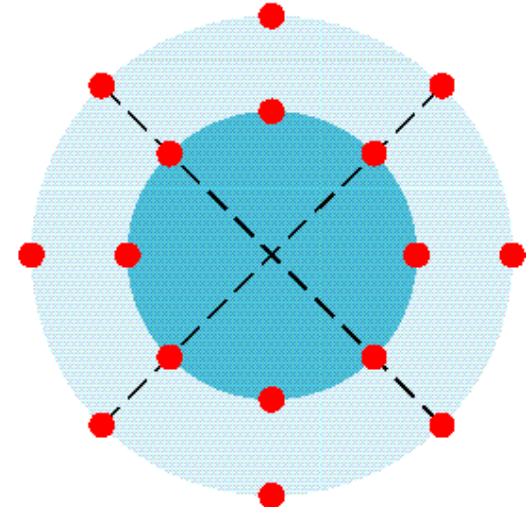
16-QAM

4 amplitudes,
8 phases



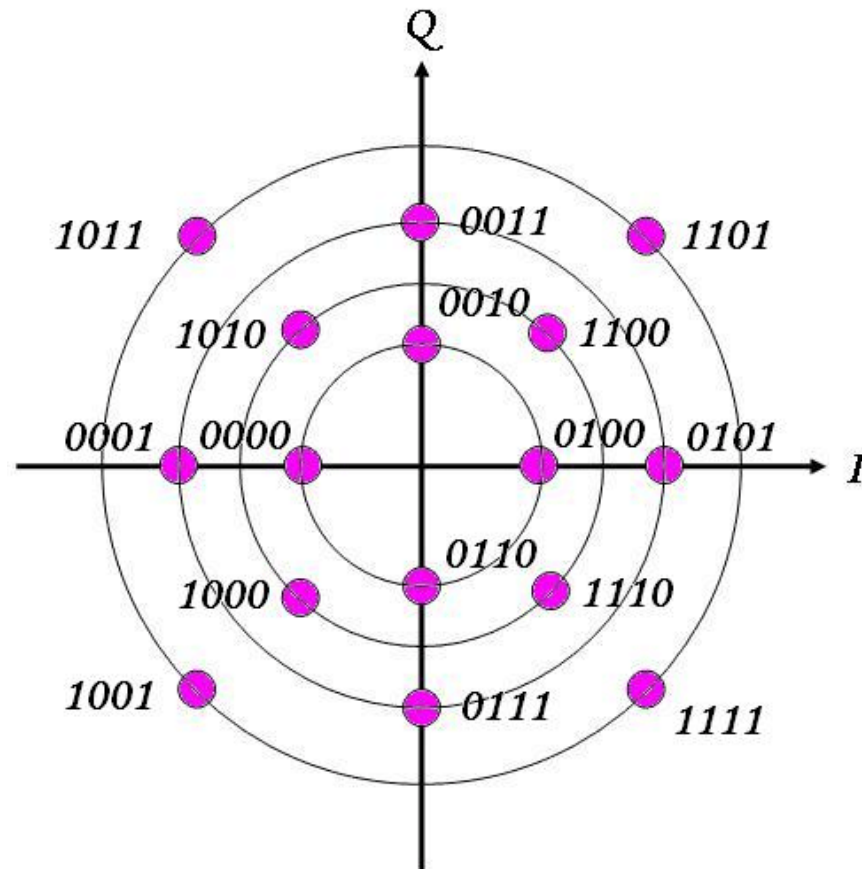
16-QAM

2 amplitudes,
8 phases



16-QAM

16-QAM Constellation

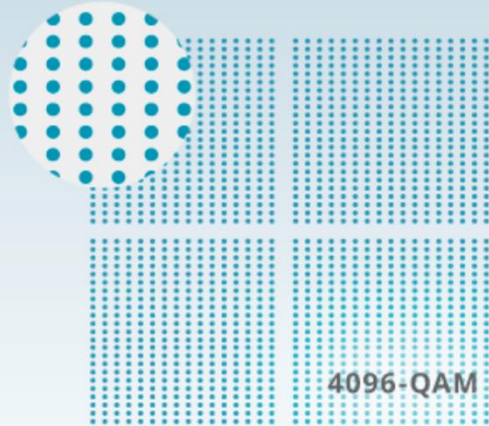


WiFi 7

Wi-Fi 7

12 bits/symbol

802.11be

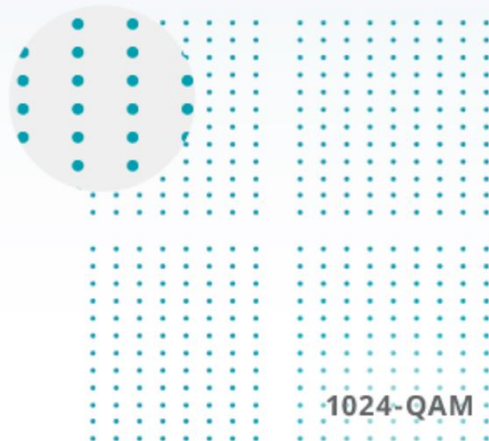


Wi-Fi 7
4096-QAM

Wi-Fi 6

10 bits/symbol

802.11ax



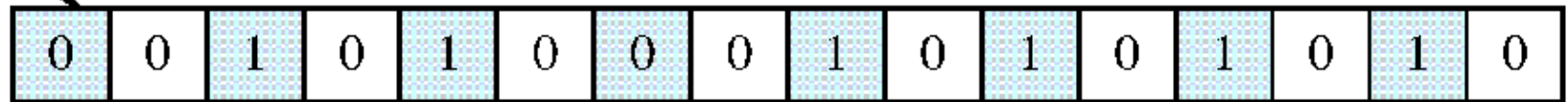
Wi-Fi 6
1024-QAM

Bit Rate and Baud Rate

Bit

Baud rate = N

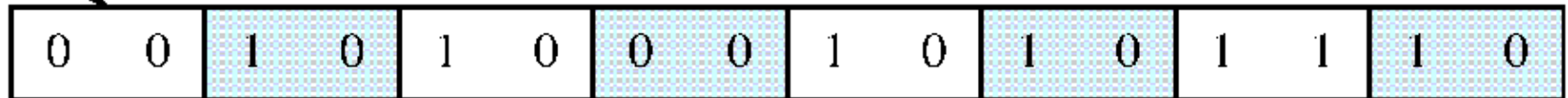
Bit rate = N



Dibit

Baud rate = N

Bit rate = $2N$



Bit Rate and Baud Rate

Tribit

Baud rate = N

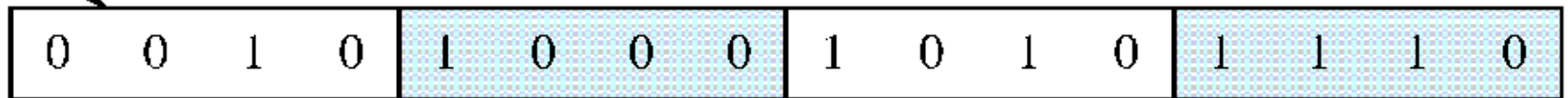
Bit rate = $3N$

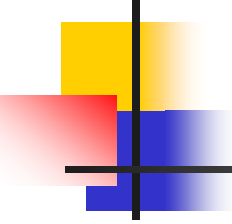


Quadbit

Baud rate = N

Bit rate = $4N$

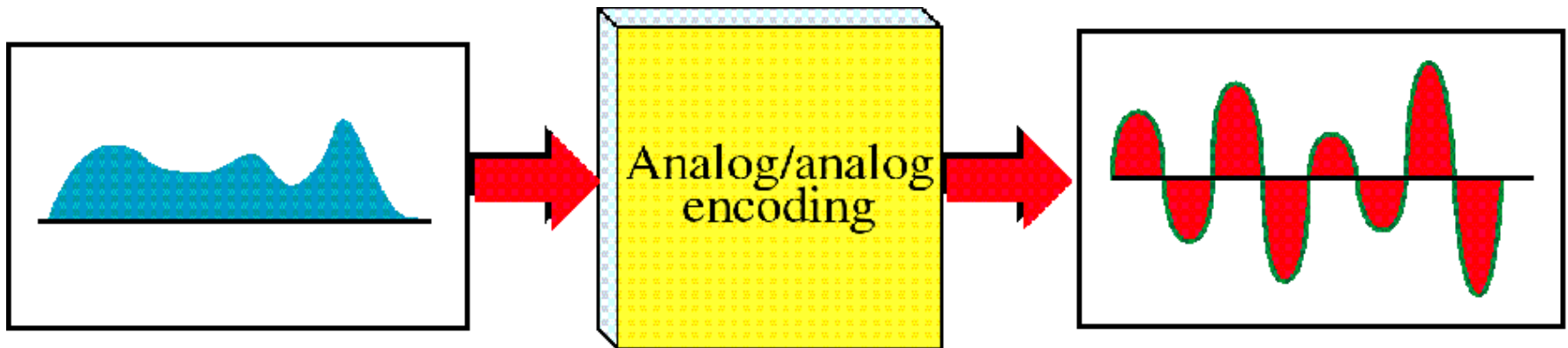




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Analog to Analog Encoding



Modulation

```
graph TD; A[Analog/Analog Encoding] --> B[AM]; A --> C[FM]; A --> D[PM];
```

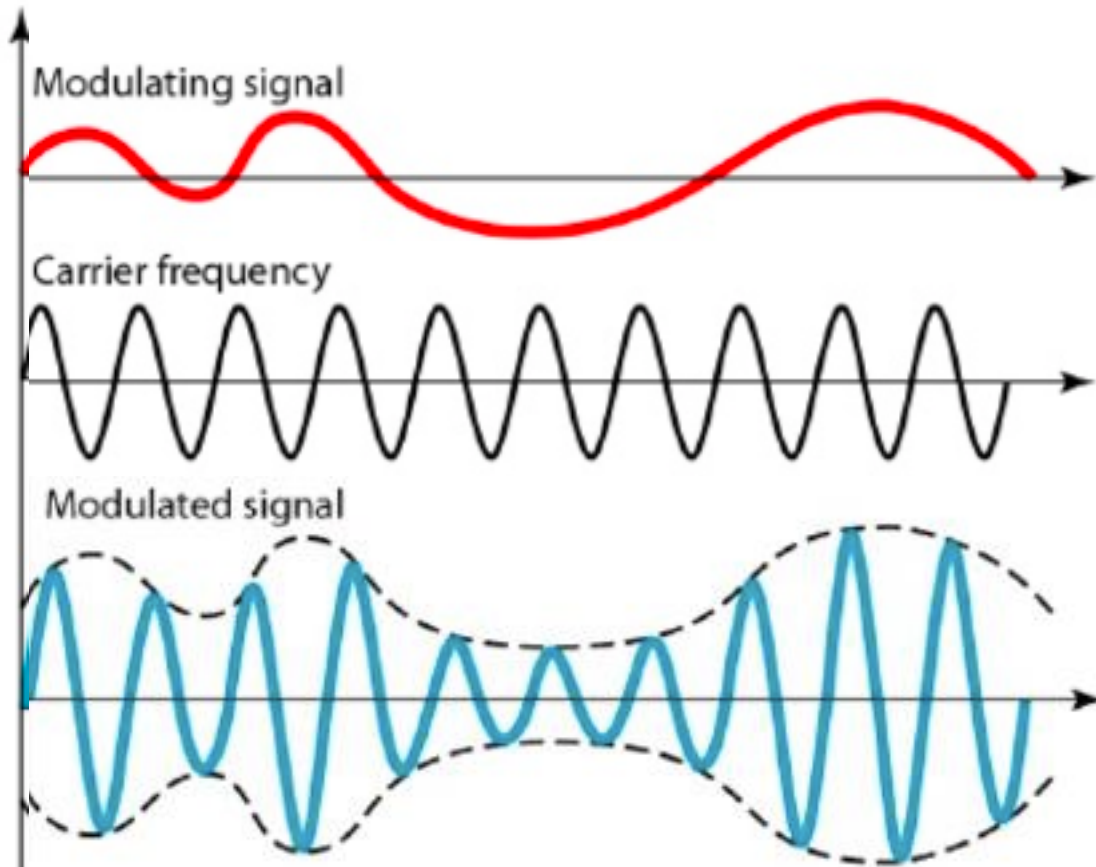
**Analog/Analog
Encoding**

AM

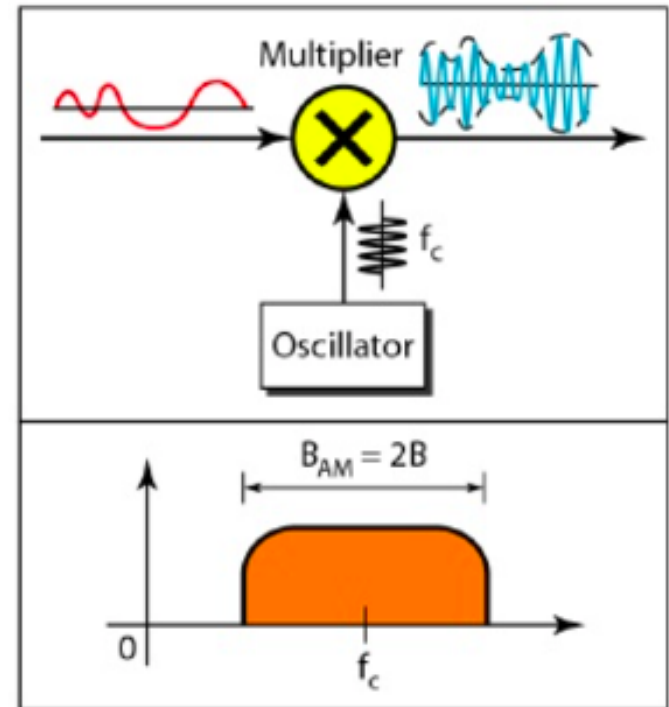
FM

PM

Amplitude Modulation



Time Domain



https://pt.slideshare.net/Akanksha_Seth/amplitude-modulation

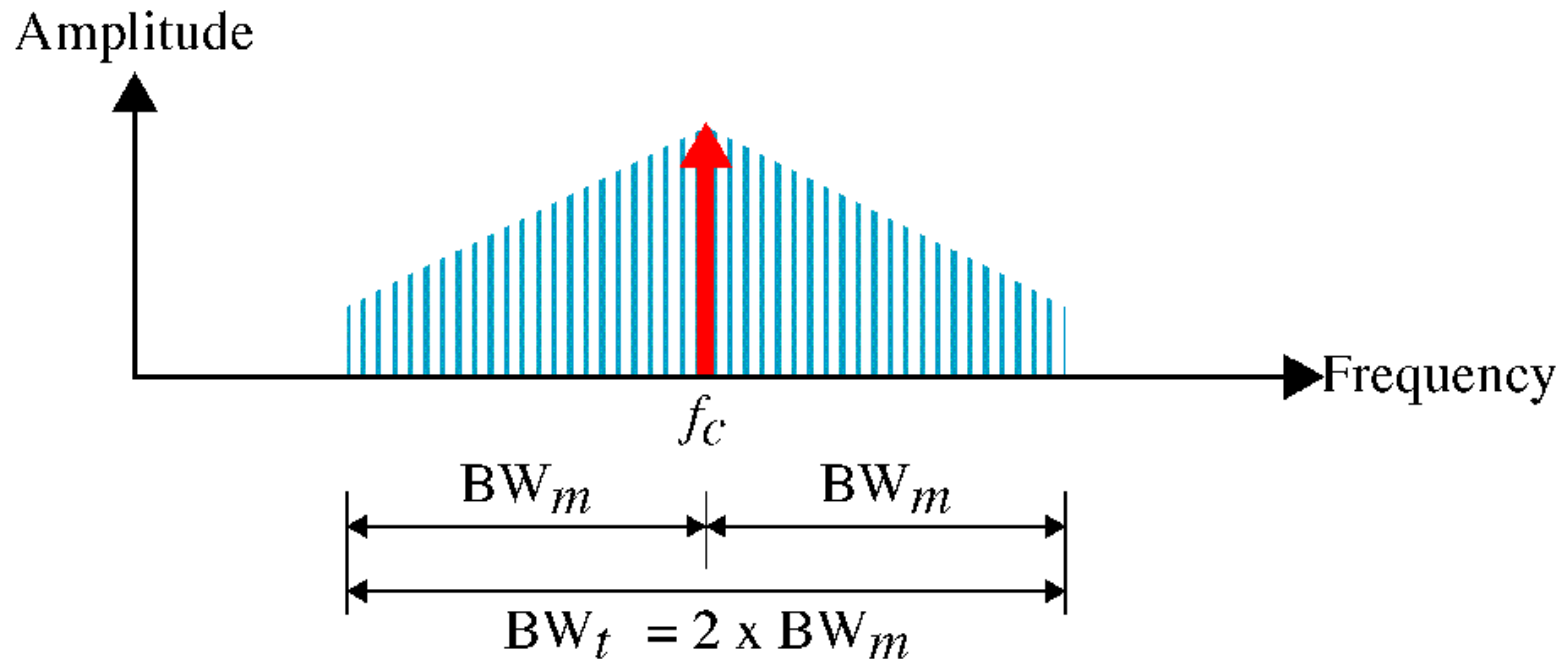
AM Bandwidth

Frequency Domain

BW_m = Bandwidth of the modulating signal (audio)

BW_t = Total bandwidth (radio)

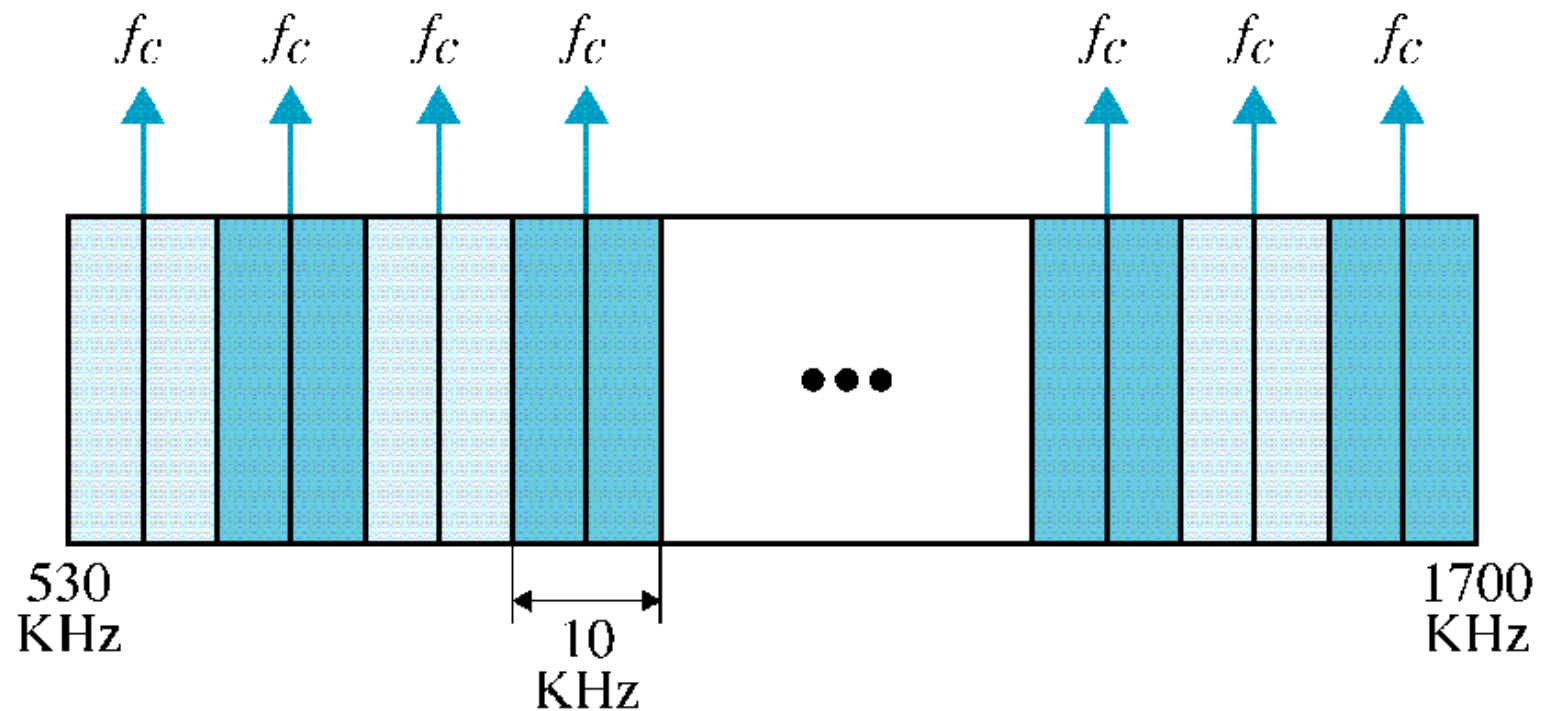
f_c = Frequency of the carrier



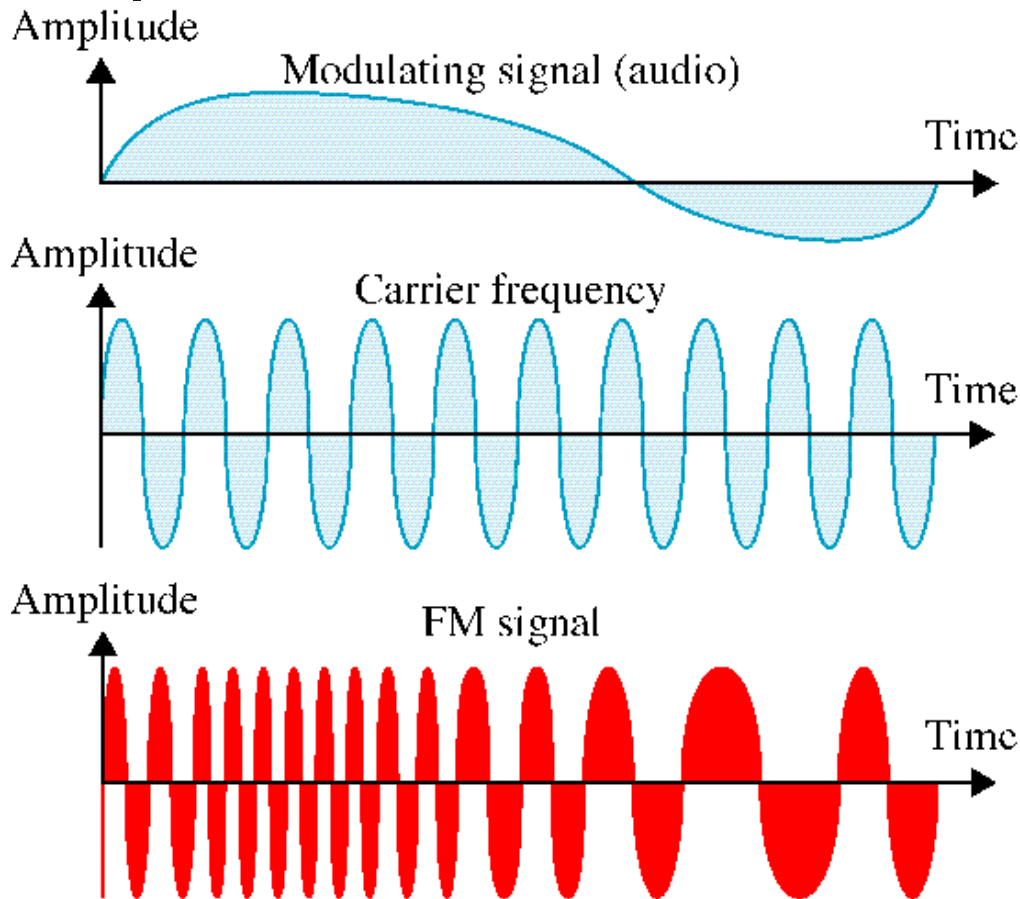
AM Band Allocation

Frequency Domain

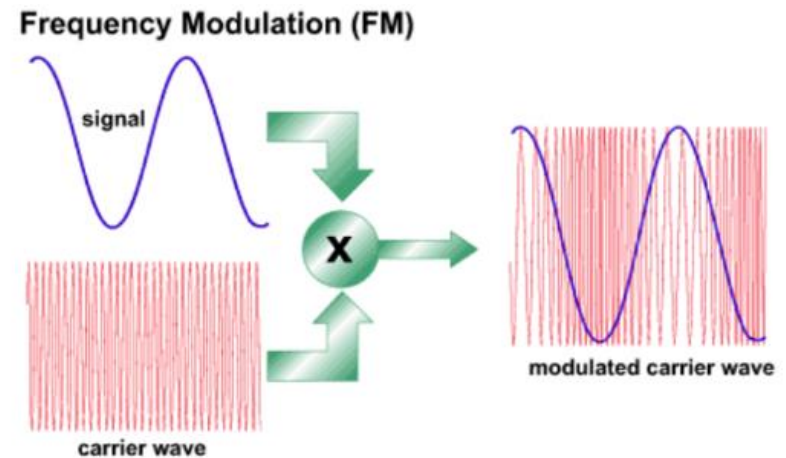
f_c = Carrier frequency of the station



Frequency Modulation



Time Domain



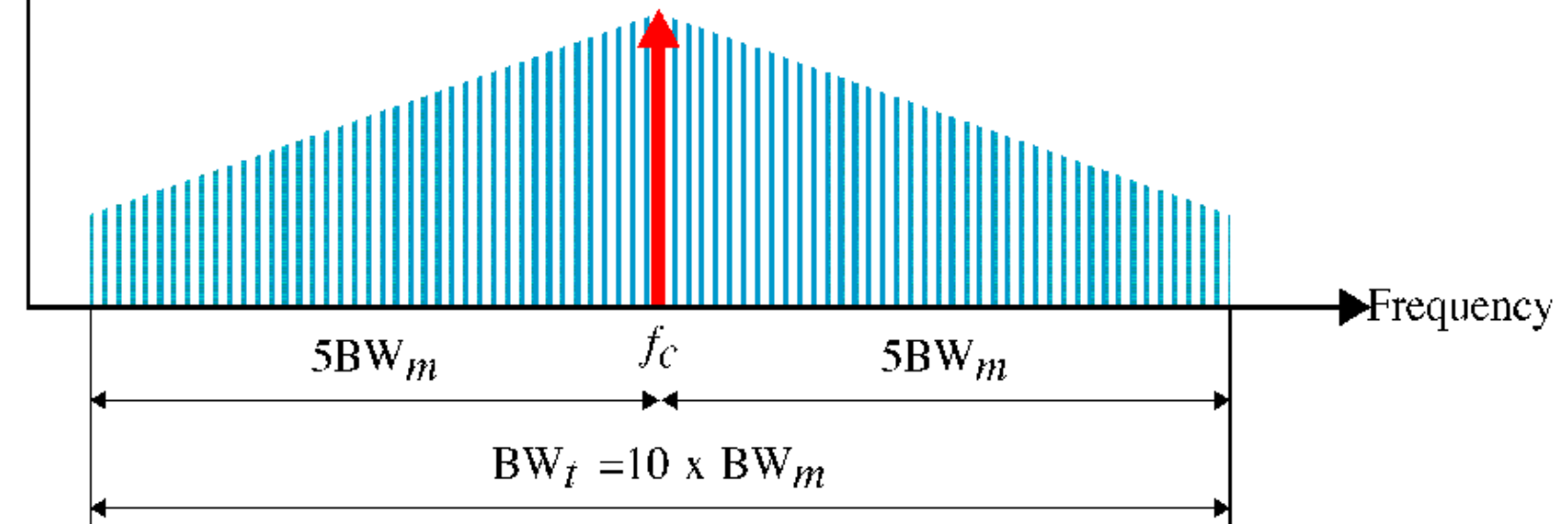
<https://medium.com/@kanyaratmtr/modulation-เป็นกระบวนการเปลี่ยนช่วงความถี่ให้เหมาะสมกับการส่งสัญญาณ-demodulation-6d9c8ce54486>

FM Bandwidth

Frequency Domain

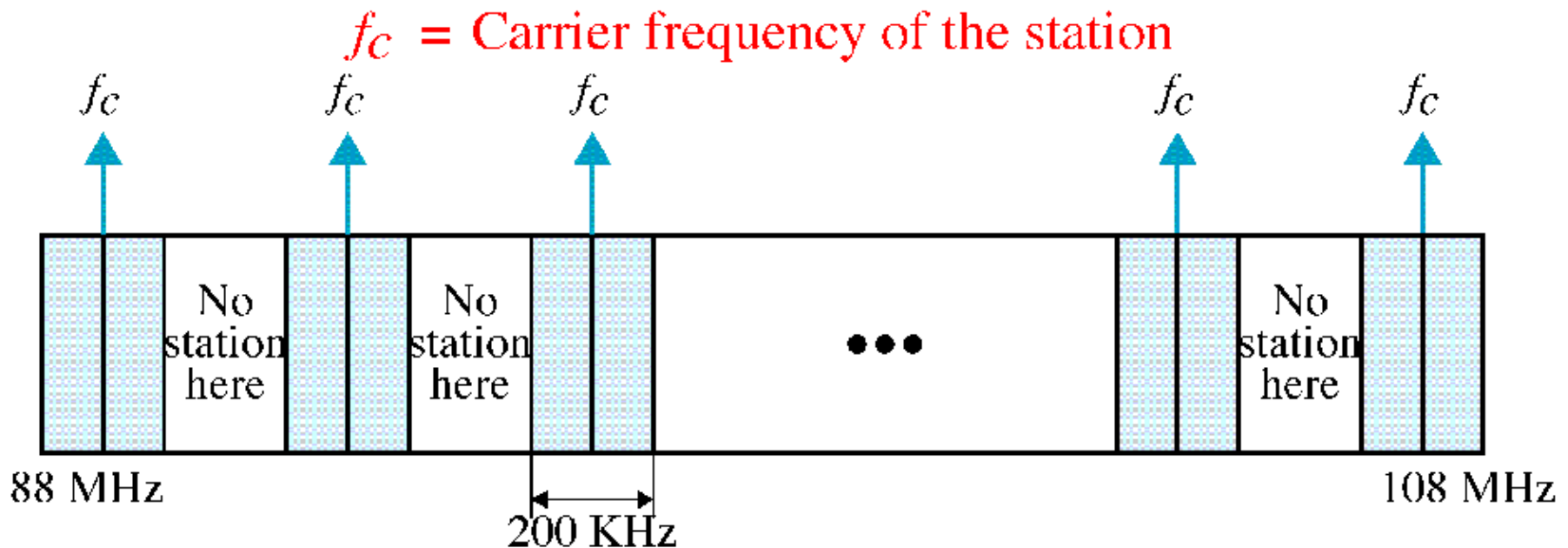
Amplitude

BW_m = Bandwidth of the modulating signal (audio)
 BW_t = Total bandwidth (radio)
 f_c = Frequency of the carrier



FM Band Allocation

Frequency Domain



Audio stereo = 15KHz → 150KHz

FCC → 200 KHz

Phase Modulation

